CODE-IF: A Convex/Deep Image Fusion Algorithm for Efficient Hyperspectral Super-Resolution

Chia-Hsiang Lin, *Member, IEEE*, Cheng-Ying Hsieh, *Student Member, IEEE*, and Jhao-Ting Lin, *Student Member, IEEE*

***Abstract*— Super-resolving remotely acquired hyperspectral images (HSIs), often with low resolution (LR), is a critical signal-processing technique, as it greatly affects the subsequent material classification and identification tasks. An economical approach in the remote sensing area is to fuse the spatial details extracted from the high-resolution (HR) counterpart multispectral image (MSI) into the LR HSI, thereby inferring the desired HR HSI. Convex analysis is an effective tool for the fusion mission, but it often relies on sophisticated regularization schemes to tackle this challenging inverse problem. In the existing literature, the deep plug-and-play (PnP) strategy was proposed for fast implementation of those sophisticated regularizers, but just approximately without convergence guarantees. Thus, we introduce deep learning (in an alternative approach) to tailor a simple convex regularizer for efficient super-resolution. Remarkably, though typical deep fusion methods can tackle nonlinear effects presented in real hyperspectral data, they often rely on big data and sophisticated network structures, which are often time-consuming and resource-intensive. Instead, our deep regularizer just needs a small-data-driven simple network architecture that implies better stability and tractability; we achieve so by reconsidering the role of deep learning as simply to guide the convex algorithm to search the fusion solution, rather than directly serving as the final solution. The proposed convex/deep image fusion (CODE-IF) algorithm, with all the closed-form algorithmic expressions derived, achieves state-of-the-art hyperspectral super-resolution (HSR) performance.**

***Index Terms*— Convex optimization (CO), deep learning (DE), hyperspectral image (HSI), image fusion, image super-resolution.**

I. Introduction

In the space exploration era, super-resolving remotely acquired hyperspectral images (HSIs), which are often with low resolution (LR), is a crucial signal-processing technique. It significantly contributes to the improvement of subsequent material classification and identification tasks. In the obtained high-resolution (HR) HSI, the HR pixels contribute to accurate material classification, while the densely populated hyperspectral bands contribute to highly effective material identification [1]. Note that spatial resolution of HSIs is often more limited than multispectral images (MSIs), because hyperspectral imager splits the incident light into more densely populated frequency bands, making each pixel covering a larger spatial region for collecting sufficient photons to achieve a satisfactory signal-to-noise ratio [2], [3]. Under such a hardware limitation, an economical approach in the remote sensing area is to fuse the spatial details extracted from the HR counterpart MSI (covering the same spatial region) into the LR HSI, thereby inferring the desired HR HSI [4], [5]. In addition, we remark that acquiring counterpart HSIs/MSIs under similar atmospheric and illumination conditions becomes increasingly feasible due to the ever-improving revisiting cycles of the hyperspectral/multispectral satellites [6].

Convex optimization (CO), which has demonstrated significant success in miniaturized satellite imaging [7], also serves as a powerful mathematical tool for super-resolution tasks. One of the notable CO-based methods is the hyperspectral super-resolution algorithm (HySure) [8] that adopts the popular total-variation (TV) regularizer to get piecewise-smooth results. However, the nonquadratic and nonsmooth nature of TV impedes the real-time applicability of HySure. Another fusion method, the CO-based coupled nonnegative matrix factorization (CO-CNMF) algorithm [3], employs the convex sum-of-squared-distance (SSD) regularizer to minimize the volume of the hyperspectral signature simplex. Nevertheless, the original form of the minimization problem is NP-hard [9]. Therefore, CO-based fusion methods, though powerful, often rely on intricate regularization schemes to achieve outstanding performance for the challenging fusion inverse problem.

In the existing literature, the above issue is often addressed by incorporating deep learning (DE). Specifically, those hardly tractable subproblems induced by mathematically sophisticated regularization are often solved by calling some DE-based denoiser—the well-known deep plug-and-play (PnP) strategy [10]. To understand how deep PnP works, note that the regularization function (embedded in some CO formulation) frequently induces a proximal operator under benchmark CO solvers (e.g., alternating direction method of multipliers (ADMM) solver [11]). Nevertheless, the regularization function is typically too complex to be explicitly expressed, making its corresponding proximal operator unsolvable under a closed-form setting, so deep PnP is introduced to implement this

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proximal operator. Such a proximal operator can be physically explained as a denoiser [10], which yields a clean image with desired properties (encouraged by the adopted regularization scheme) to be used as an *approximated* solution to the aforementioned hard subproblem. For instance, CNN-Fus [12] is a method based on the subspace representation and convolutional neural network (CNN) denoiser, where the CNN is well-trained for gray-level image denoising. While deep PnP appears to be promising for addressing the above issue, the induced proximal operator is just *approximately* implemented by DE, meaning that the deep PnP methods do not have algorithmic convergence guarantees in general.

On the other hand, DE is known to capture nonlinear effects commonly observed in real remote sensing data, bringing an advantage over pure CO-based fusion algorithms [13]. Also, DE enables automatic updates of all parameters in the network during the training phase, allowing the network to capture implicit priors, and this, in fact, leads to several outstanding DE-based fusion methods including multiscale and multidepth CNN (MSDCNN) [14], residual two-stream fusion network (ResTFNet) [15], spatial–spectral reconstruction network (SSR-Net) [16], and dual-structure U-Net (D-UNet) [17]. DE-based methods often rely on complex architectures with million-scale parameters. To ensure that the trained deep neural network can generate a reasonably good solution, these methods, including deep PnP strategies, often require a substantial amount of training data (big data). However, a key challenge lies in the limited availability of hyperspectral data compared to the huge amount of RGB data available everywhere on the internet, making big data collection often costly, time-consuming, and even unavailable in practical scenarios.

To against the aforementioned problems, we propose a convex/deep image fusion (CODE-IF) algorithm equipped with a small-data-driven DE technique to achieve mathematically simple convex regularization (e.g., quadratic). By reconsidering the role of the DE technique as simply guiding the convex algorithm to search for the fusion solution, rather than directly serving as the final solution, CODE-IF achieves outstanding fusion performance using just a small-data-driven simple network architecture. Specifically, we employ a simple convex regularizer, the $\boldsymbol{Q}$-quadratic function, to extract valuable information from the DE solution, which may be weak but sufficient to guide the algorithm in completing challenging satellite remote sensing tasks [18]. Remarkably, the proposed CODE-IF is not merely a conventional two-stage process (see recent works [19], [20]), but similar tricks can assist software developers in avoiding sophisticated optimization procedures in their future projects, just like CODE-IF that avoids using math-heavy regularizers in the criterion design. The main contributions of this article are outlined below.

1) In this article, both closed-form algorithmic expressions and convergence guarantees will be built for the CODE-IF algorithm, whose state-of-the-art performance will be demonstrated through extensive experiments.
2) Under the CODE framework, we demonstrate how to avoid math-heavy optimization procedures by introducing a simple $\boldsymbol{Q}$-quadratic regularization in the CO part (leading to a concise and efficient algorithm), while avoiding big data collection in the DE part.
3) This article has successfully applied the elegant CODE theory to the critical hyperspectral super-resolution (HSR) problem for the first time. This well demonstrates that small-data machine learning is sufficient to achieve state-of-the-art performance even for challenging inverse problems, including metasurface-driven snapshot hyperspectral imaging [21], and this is important for remote sensing as the acquisition of a sufficient amount of high-quality hyperspectral data is often costly.
4) CODE-IF does not rely on advanced DE network structure because a rough solution, returned by a naive DE network, is sufficient for effective regularization, thanks to the judicious design of $\boldsymbol{Q}$-quadratic feature extractor customized for hyperspectral data. In other words, CODE-IF is easy to use for a broader range of software developers, rather than manageable only to experienced DE experts, making CODE-IF user-friendly.

The remaining parts of this article are organized as follows. The related works about the HSR problem will be reviewed in Section II. In Section III, the main idea of the proposed CODE-IF will be illustrated in detail. Comprehensive experimental investigations and discussions will then be presented in Section IV, and concluding remarks will be drawn in Section V. Finally, to avoid corrupting the reading smoothness, mathematical details, and proofs will be collectively given in the Appendix.

For better readability, some standard notations used in this article are defined hereinafter. $\mathbb{R}^n$ is the n-dimensional Euclidean space, and $\mathbb{R}^{m\times n}$ is the set of all the real-valued $m\times n$ matrices. $\mathbf{I}_n$ denotes the $n\times n$ identity matrix. $\|\cdot\|_F$ and $\|\cdot\|_2$ stand for the Frobenius norm and ℓ_2-norm, respectively. The notation "$\otimes$" denotes the Kronecker product. Given a matrix $\mathbf{X}\in\mathbb{R}^{m\times n}$, $\mathrm{vec}(\mathbf{X})\in\mathbb{R}^{mn}$ denotes its vectorization, $\mathrm{span}(\mathbf{X})$ denotes the subspace spanned by the column vectors of $\mathbf{X}$, and $\mathbf{X}^T$ is its transpose. Finally, $\mathbf{U}^{-1}$ denotes the inverse of the given square matrix $\mathbf{U}\in\mathbb{R}^{n\times n}$.

II. Related Works

In this section, some fusion methods will be reviewed, and they can be categorized into traditional ones and DE-based ones, which are presented in Sections II-A and II-B, respectively. Additionally, we will select some latest methods to analyze their strengths and weaknesses. To solve the challenging HSR problem, we employ the CODE theory [18], which will be concisely recalled in Section II-C for self-contained purposes.

A. Traditional Methods

Recently, CO-based methods have exhibited their efficacy as valuable mathematical tools for HSI/MSI fusion. Typically, one category in the CO-based methods uses tensor factorization; for instance, the low tensor multi-rank (LTMR) regularization method [22] imposes the LTMR prior to exploit the nonlocal similarities. Also, the nonlocal similarities are utilized in [23] and [24]. Although the objective function

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is nonconvex (i.e., no convergence guarantee) [23], and the Bayesian inference may not be consistent with the real-world scene [24], they still can demonstrate the advantages by extensive experiments. Another noteworthy category in CO-based methods decomposes a target matrix into its spectrally and spatially degraded versions. For example, the HySure algorithm [8] addresses fusion using a convex objective function that includes two quadratic data-fitting terms and a convex vector-TV (VTV) regularization term. The VTV term promotes piecewise-smooth fusion solutions with spatial discontinuities aligned across different hyperspectral bands. Besides, we must also mention the CO-CNMF algorithm [3]. In CO-CNMF, the nonconvex simplex volume regularization function is trickily approximated using the convex SSD function, where SSD is designed to minimize the distances between the underlying hyperspectral signature vectors, thereby reducing the corresponding volume of the signature simplex. While this approach is theoretically well-grounded under the rather mild and practical condition of $\gamma > 1/(N-1)^{1/2}$ (γ is the pixel purity and N is the number of endmembers) [25], its original form induces a hardly tractable NP-hard optimization procedure [9]. Similarly, the VTV regularizer mentioned earlier also hinders real-time applicability in HySure, primarily due to its nonquadratic and nonsmooth regularization nature [8]. As a result, CO-based fusion methods frequently rely on regularization strategies.

B. DE-Based Methods

DE-based methods have shown their outstanding capability of capturing nonlinear effects in various inverse imaging tasks. For instance, MSDCNN [14] employs filters of different sizes to extract features at varying scales. In contrast to most networks that stack upsampled LR HSIs and HR MSIs as inputs, ResTFNet [15] utilizes a two-stream encoder to extract information independently from LR HSIs and HR MSIs. Gao et al. [26] proposed an unmixing-based method [27] that decomposes the input data into two modalities (i.e., abundances and endmembers). Despite having good performance, the key parameter of endmember number cannot be automatically determined. In D-UNet [17], two U-Nets are integrated. The first U-Net uses an encoder–decoder architecture to extract features of various shapes and sizes from HR MSIs, while the second U-Net employs multiscale feature extraction along with a spatial–spectral attention scheme. As DE-based methods often involve complicated architectures and result in expensive computation and storage costs, HSRnet [16] incorporated attention modules to effectively extract data without too much additional burden. Similarly, Ran et al. [28] applied a recursive mechanism also for reducing the computational burden. Although spectrum information may be underutilized, the spatial details are regarded as a guiding term and the method also achieves excellent performance in various resolution enhancement tasks. As most DE-based methods mentioned above are empirically designed, which are not explainable (just like a "black box"), the spatial–spectral reconstruction network (SSR-Net) [29] attempts to construct an interpretable network by incorporating spatial/spectral edge loss. Furthermore, through the decomposition of an iterative optimization algorithm into a hierarchical network structure, deep unrolling methods such as the MSI/HSI fusion network (MHFnet) [30], [31] and model-guided deep convolutional network [32] show impressive interpretability and generalization ability.

Apart from the mentioned supervised methods, we should review some unsupervised methods. As an extension of the deep image prior (DIP) [33], Uezato et al. [34] proposed a guided deep decoder to provide better regularization for the fusion problems, and Fu et al. [19] utilizes an externally pretrained network as a deep prior and then unsupervised trains it from a single image. Also, Zhang et al. [35] designed a complicated pretrained fusion module to obtain roughly fused HSI and exploited the image-specific statistics by training a simple degeneration network. Incorporating the concept of matrix decomposition and physical properties, Qu et al. [36] enforced the low-dimensional basis to follow a Dirichlet distribution, while Zheng et al. [37] constrained the sum of abundances to meet the sum-to-one nature. Moreover, there are extended versions available to tackle the misalignment fusion problem [36], [38] or adaptively learn the spectral responses [38], [39]. The impressive performances of these methods allude to the necessity of incorporating DE to assist CO-based fusion.

C. CODE Theory for Solving Inverse Problems

Considering that CO and DE are two of the most popular frameworks for solving inverse problems, but have their own disadvantages in practice, a new theoretical framework is desired to rule out those disadvantages. Specifically, the CODE framework, invented in recent remote sensing literature [18], trickily bypasses the big data collection (often required in DE) and the math-heavy regularization scheme (often required in CO), where the former is usually expensive and time-consuming, while the latter would lead to a high-computational-complexity algorithm. The ADMM optimizer in CO [11] and the adaptive moment estimation (Adam) optimizer in DE [40] are judiciously blended to form the CODE theory.

To see how the CODE theory avoids math-heavy regularization even in the absence of advanced DE networks and big data, we recall the following convex learning criterion typically adopted for solving inverse problems:

$$\boldsymbol{x}^\star = \arg\min_{\boldsymbol{x}\in\mathcal{S}} \ \mathrm{DF}(\boldsymbol{x}) + \lambda\,\mathrm{REG}(\boldsymbol{x}) \tag{1}$$

where the constraint set $\mathcal{S}$ forces some natural properties to the target image $\boldsymbol{x}$, $\mathrm{DF}(\boldsymbol{x})$ is the data fidelity term for yielding a reasonable solution $\boldsymbol{x}^\star$, and $\mathrm{REG}(\boldsymbol{x})$ is the regularizer for encouraging some desired physical properties (e.g., sparsity) in the final solution $\boldsymbol{x}^\star$ with the regularization strength controlled by $\lambda > 0$.

The success of CODE lies within the judicious design of a simple but effective convex regularization scheme. To see it, notice that those most commonly used regularizers are often nondifferentiable (e.g., TV regularizer, or ℓ_1 regularizer) or complicated (e.g., the graph-embedded self-similarity (SS) regularizer [41]). By contrast, the regularizer proposed in CODE [18] has a very simple and differentiable math form,

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thereby leading to a computationally very efficient inverse imaging algorithm for solving (1). To be specific, the CODE theory adopts the following convex regularization scheme, which can be regarded as a generalized version of the Tikhonov regularization [42], that is,

$$\mathrm{REG}(\boldsymbol{x}) \triangleq \frac{1}{2}\|\boldsymbol{x}-\boldsymbol{x}_{\mathrm{DE}}\|_{\boldsymbol{Q}}^2 \tag{2}$$

where $\boldsymbol{x}_{\mathrm{DE}}$ is an intermediate solution obtained by some DE technique and $\|\cdot\|_{\boldsymbol{Q}}$ denotes the $\boldsymbol{Q}$-quadratic norm, defined as $\|\boldsymbol{v}\|_{\boldsymbol{Q}}^2 \triangleq \boldsymbol{v}^T\boldsymbol{Q}\boldsymbol{v}$ for a given vector $\boldsymbol{v}$ and a positive semidefinite (PSD) matrix $\boldsymbol{Q}$ [43]. In [18], the PSD matrix is designed based on the natural phenomenon that hyperspectral data is usually located in a low-dimensional subspace. We remark that though TV and SS are also convex regularizers, they have more complicated mathematical forms often resulting in math-heavy optimization procedures in the final implementation.

Another great advantage of CODE is that the intermediate DE solution $\boldsymbol{x}_{\mathrm{DE}}$ is allowed just to be a roughly estimated weak solution (see [18, Fig. 1]) because it has been proven that the simple regularizer (2) is capable of extracting useful information (to be fused into the final solution $\boldsymbol{x}^\star$) from a weak solution $\boldsymbol{x}_{\mathrm{DE}}$ learned from small data. The rough guess $\boldsymbol{x}_{\mathrm{DE}}$ to the final solution $\boldsymbol{x}^\star$ can be obtained by training a naive DE network without big data collection or experienced architecture design. To see how this would be possible, we take the TV regularizer for example, which just *indirectly* encourages edge-preserving solutions $\boldsymbol{x}^\star$ without knowing the location of edges ahead of time. By contrast, even if $\boldsymbol{x}_{\mathrm{DE}}$ is a weak solution, it still contains useful texture information learned from small data (see [18, Fig. 1]), hence *directly* promoting an edge-preserving solution $\boldsymbol{x}^\star$ with better prior knowledge on the edge location, and in this regard, the simple regularizer (2) is actually more effective than the prestigious TV regularizer. Therefore, as long as the roughly guessed solution $\boldsymbol{x}_{\mathrm{DE}}$ contains useful property/structure, it is helpful for the subsequent regularization task.

Below, we recall some basics of convex analysis. Once the convex criterion (1) has been properly designed, the CODE framework effectively solves (1) using ADMM, in which the augmented Lagrangian in the dual space can be derived as

$$\mathcal{L}(\boldsymbol{x},\boldsymbol{z},\boldsymbol{d}) \triangleq \mathrm{DF}(\boldsymbol{x}) + \lambda\,\mathrm{REG}(\boldsymbol{z}) + \frac{p}{2}\|\boldsymbol{x}-\boldsymbol{z}-\boldsymbol{d}\|_2^2 \tag{3}$$

where $\boldsymbol{d}$ is the scaled dual variable, $\boldsymbol{z}$ is a newly introduced primal variable for reformulating (1) into the ADMM-form [11], and the penalty parameter is empirically set as $p := 0.1$. ADMM is computationally efficient, particularly for large-scale optimization problems, as its dual decomposability simplifies complex optimization problems into a series of relatively simple subproblems [44]. Specifically, at the $(q+1)$th iteration, two convex subproblems are sequentially updated by

$$\begin{aligned}\boldsymbol{z}^{q+1} &:= \arg\min_{\boldsymbol{z}\in\mathcal{S}} \mathcal{L}\left(\boldsymbol{x}^q,\boldsymbol{z},\boldsymbol{d}^q\right)\\ &= \arg\min_{\boldsymbol{z}\in\mathcal{S}} \lambda\,\mathrm{REG}(\boldsymbol{z}) + \frac{p}{2}\|\boldsymbol{x}^q-\boldsymbol{z}-\boldsymbol{d}^q\|_2^2 \qquad (4)\\ \boldsymbol{x}^{q+1} &:= \arg\min_{\boldsymbol{x}\in\mathcal{S}} \mathcal{L}\left(\boldsymbol{x},\boldsymbol{z}^{q+1},\boldsymbol{d}^q\right)\\ &= \arg\min_{\boldsymbol{x}\in\mathcal{S}} \mathrm{DF}(\boldsymbol{x}) + \frac{p}{2}\|\boldsymbol{x}-\boldsymbol{z}^{q+1}-\boldsymbol{d}^q\|_2^2. \qquad (5)\end{aligned}$$

Algorithm 1 Alternating Direction Method of Multipliers [11]

1: **Given** the penalty parameter $p > 0$ and the small data solution $\boldsymbol{x}_{\mathrm{DE}}$. Initialize $q := 0$, $\boldsymbol{d}^q$, and $\boldsymbol{x}^q$.
2: **repeat**
3: update $\boldsymbol{z}^{q+1}$ by solving Eq. (4);
4: update $\boldsymbol{x}^{q+1}$ by solving Eq. (5);
5: update $\boldsymbol{d}^{q+1} := \boldsymbol{d}^q - (\boldsymbol{x}^{q+1}-\boldsymbol{z}^{q+1})$;
6: $q := q+1$;
7: **until** the predefined stopping criterion is met.
8: **Output** $\boldsymbol{x}^\star := \boldsymbol{x}^q$.

Then, $\boldsymbol{d}$ is updated using the gradient method, leading to the ADMM algorithm as summarized in Algorithm 1.

For interested readers who wonder how such a simple CODE theory can effectively solve challenging inverse problems in hyperspectral remote sensing, comprehensive discussions from both theoretical and experimental perspectives have been presented in [18, Sec. II.B]. In this work, without the aid of big data, we employ the promising CODE theoretical framework to solve the fusion problem for the first time, achieving state-of-the-art HSR performance.

III. CODE-Based HSI/MSI Fusion Algorithm

In Section III-A, we will recall the signal model for the CO-based HSI/MSI fusion and accordingly define a convex fusion criterion with deep regularization. Section III-B presents the detailed design of a simple deep model, as well as a refined low-complexity convex algorithm. Finally, we will give a brief discussion in Section III-C.

A. Convex Eigenspace-Based HSR Criterion Design

To solve the inverse problem, we begin by describing the corresponding forward model below, as graphically illustrated in Fig. 1. To super-resolve the observed LR HSI $\mathbf{Y}_h \in \mathbb{R}^{M\times L_h}$ to obtain the target HR HSI $\mathbf{Z} \in \mathbb{R}^{M\times L}$, an economical and practical approach is to fuse the spatial information embedded in the counterpart HR MSI $\mathbf{Y}_m \in \mathbb{R}^{M_m\times L}$ into $\mathbf{Y}_h$. Here, M_m denotes the number of multispectral bands in $\mathbf{Y}_m$ that is much smaller than the number M of hyperspectral bands in $\mathbf{Y}_h$, and L is the number of HR pixels in $\mathbf{Y}_m$ that is much larger than the number L_h of LR pixels in $\mathbf{Y}_h$. Thus, the desired target image $\mathbf{Z} \in \mathbb{R}^{M\times L}$, and the observable quantities $\mathbf{Y}_m$ and $\mathbf{Y}_h$ can be, respectively, described as spectrally degraded (i.e., $M_m < M$) and spatially blurred (i.e., $L_h < L$) versions of $\mathbf{Z}$.

Therefore, by letting $\mathbf{D} \in \mathbb{R}^{M_m\times M}$ and $\mathbf{B} \in \mathbb{R}^{L\times(L/r^2)}$ be the spectral response transform matrix and the spatial spread transform matrix, respectively, we have the following forward model, that is,

$$\mathbf{Y}_m = \mathbf{D}\mathbf{Z} \tag{6}$$

$$\mathbf{Y}_h = \mathbf{Z}\mathbf{B} \tag{7}$$

where $\mathbf{D}$ spectrally downsamples $\mathbf{Z}$ to yield the MSI $\mathbf{Y}_m$, and $\mathbf{B}$ spatially blurs $\mathbf{Z}$ to yield the LR image $\mathbf{Y}_h$. In this work, we assume that the blurring factor $r \triangleq (L/L_h)^{1/2}$ is an integer and that $\mathbf{B}$ and $\mathbf{D}$ are assumed to be known *a priori* as

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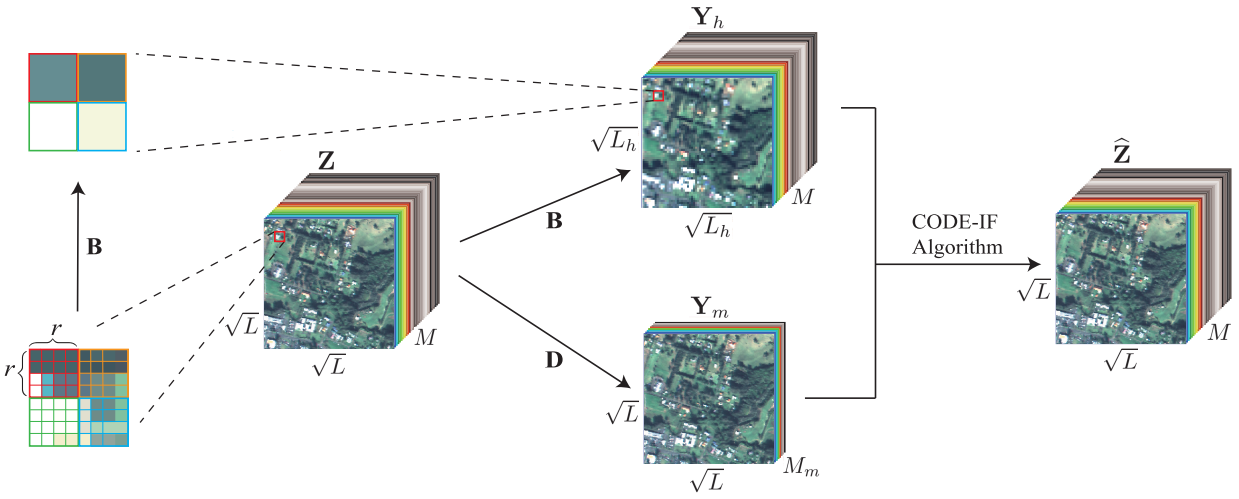


Fig. 1. Graphical illustration of the HSR problem, where the HR MSI $\mathbf{Y}_m$ and the LR HSI $\mathbf{Y}_h$ are fused to obtain the target HR HSI $\mathbf{Z}$. The observable $\mathbf{Y}_m$ (resp., $\mathbf{Y}_h$) can be modeled as a spectrally downsampled (resp., spatially blurred) version of $\mathbf{Z}$. Each LR pixel in $\mathbf{Y}_h$ corresponds to a spatial region containing r^2 HR pixels in $\mathbf{Z}$, or, in other words, $\mathbf{Z}$ is regarded as the super-resolved version of $\mathbf{Y}_h$ via the proposed CODE-IF algorithm.

they can be obtained by image registration and radiometric calibration [3], [45], [46]. Therefore, a natural data-fitting criterion can be cast as

$$\mathrm{DF}(\mathbf{Z}) = \|\mathbf{Y}_m - \mathbf{D}\mathbf{Z}\|_F^2 + \|\mathbf{Y}_h - \mathbf{Z}\mathbf{B}\|_F^2 \tag{8}$$

which is convex but involving too many optimization variables in $\mathbf{Z} \in \mathbb{R}^{M\times L}$ as M and L are large numbers.

Considering the algorithmic stability, we need to further reduce the number of optimization variables in the data-fitting criterion. To this end, we will reformulate the data-fitting term in the so-called eigenspace [41]. To see it, note that each hyperspectral pixel in $\mathbf{Z} = [\mathbf{z}[1], \ldots, \mathbf{z}[L]]$ can be modeled as a linear combination of the signature vectors (a.k.a. endmembers) corresponding to the materials presented in the studied area [47], [48], that is,

$$\mathbf{Z} = \mathbf{A}\mathbf{S}' \tag{9}$$

where the ith column of $\mathbf{A} \in \mathbb{R}^{M\times N}$ is the ith material signature, the ith row of $\mathbf{S}' \in \mathbb{R}^{N\times L}$ is the abundance map of the ith material, and N is the number of materials in the HSI $\mathbf{Z}$. By (9), there exists a semi-unitary basis matrix $\mathbf{E} \in \mathbb{R}^{M\times N}$ (i.e., $\mathbf{E}^T\mathbf{E} = \mathbf{I}_N$) such that $\mathrm{span}(\mathbf{E}) = \mathrm{span}(\mathbf{A})$. Therefore, we can rewrite the target image as

$$\mathbf{Z} = \mathbf{E}\mathbf{S} \tag{10}$$

for some coefficient matrix $\mathbf{S} \in \mathbb{R}^{N\times L}$. Obviously, the semi-unitary matrix $\mathbf{E}$ can be computed using principal component analysis (PCA).

Specifically, according to (10), by applying PCA on data $\mathbf{Z}$, the first N principal components can be used to form the N orthogonal column vectors of $\mathbf{E}$. Nevertheless, the target image $\mathbf{Z}$ is unobservable. An alternative approach is to take the first N principal components of the observable data $\mathbf{Y}_h$ (or of the computable data $\mathbf{Z}_{\mathrm{DE}}$; see Section III-B) to form the basis matrix $\mathbf{E}$, as shown below:

$$\mathbf{E} = \left[\mathbf{q}_1\left(\mathbf{Y}_h\mathbf{Y}_h^T\right), \ldots, \mathbf{q}_N\left(\mathbf{Y}_h\mathbf{Y}_h^T\right)\right] \in \mathbb{R}^{M\times N} \tag{11}$$

in which $\mathbf{q}_i(\mathbf{Y}_h\mathbf{Y}_h^T) \in \mathbb{R}^M$ is the ith principal eigenvector of $\mathbf{Y}_h\mathbf{Y}_h^T \in \mathbb{R}^{M\times M}$. This is a valid alternative as stated in Property 1, whose proof follows from [3, Eq. (34)] and [49, Assumptions (A2)–(A3)], as detailed in Appendix A.

Property 1: Assume that the endmember matrix $\mathbf{A}$ is of full column rank [49, Assumption (A2)] and that the abundance matrix $\mathbf{S}'$ is of full row rank [49, Assumption (A3)]. The subspace spanned by the first N principal components of the observable data $\mathbf{Y}_h$ is exactly the subspace spanned by the columns of $\mathbf{E}$. □

With the information of $\mathbf{E}$, the data-fitting criterion can be recast into the eigenspace, that is,

$$\mathrm{DF}(\mathbf{S}) = \|\mathbf{Y}_m - \mathbf{D}\mathbf{E}\mathbf{S}\|_F^2 + \|\mathbf{Y}_h - \mathbf{E}\mathbf{S}\mathbf{B}\|_F^2 \tag{12}$$

which just involves NL optimization variables in $\mathbf{S}$. Note that in the original space, there are $ML \gg NL$ optimization variables in $\mathbf{Z}$; in typical hyperspectral scenarios, there are hundreds of spectral bands, but just about ten materials. Note also that $\mathrm{DF}(\mathbf{S})$ is still a convex function, though with much fewer optimization variables.

According to the CODE theory (1)–(2), as well as (10), the overall HSR criterion can be designed based on the eigenspace representation, that is,

$$\mathbf{S}^\star = \arg\min_{\mathbf{S}\in\mathbb{R}^{N\times L}} \mathrm{DF}(\mathbf{S}) + \frac{\lambda}{2}\|\mathrm{vec}(\mathbf{E}\mathbf{S} - \mathbf{Z}_{\mathrm{DE}})\|_{\boldsymbol{Q}}^2 \tag{13}$$

where $\mathbf{Z}_{\mathrm{DE}}$ is the aforementioned rough deep-learning solution that will be computed in Section III-B. Unlike those widely used nonconvex or nondifferentiable regularization functions used in the HSR area (e.g., nonconvex simplex volume regularizer or nondifferentiable sparsity-promoting ℓ_1-norm regularizer) [25], [50], [51], the CODE theory adopts the $\boldsymbol{Q}$-quadratic norm regularizer that is convex and differentiable. The regularization strength is controlled by $\lambda > 0$ that is empirically set as $\lambda := 0.05$ in this work. Once (13) is solved, $\mathbf{S}^\star$ can be used to recover the HR target image as $\mathbf{Z} := \mathbf{E}\mathbf{S}^\star$. The detailed implementation of solving the convex criterion (13) will be reported in Section III-B. The above idea of CODE-IF is graphically summarized in Fig. 2, where the DE block and the CO block will be detailed in Section III-B.

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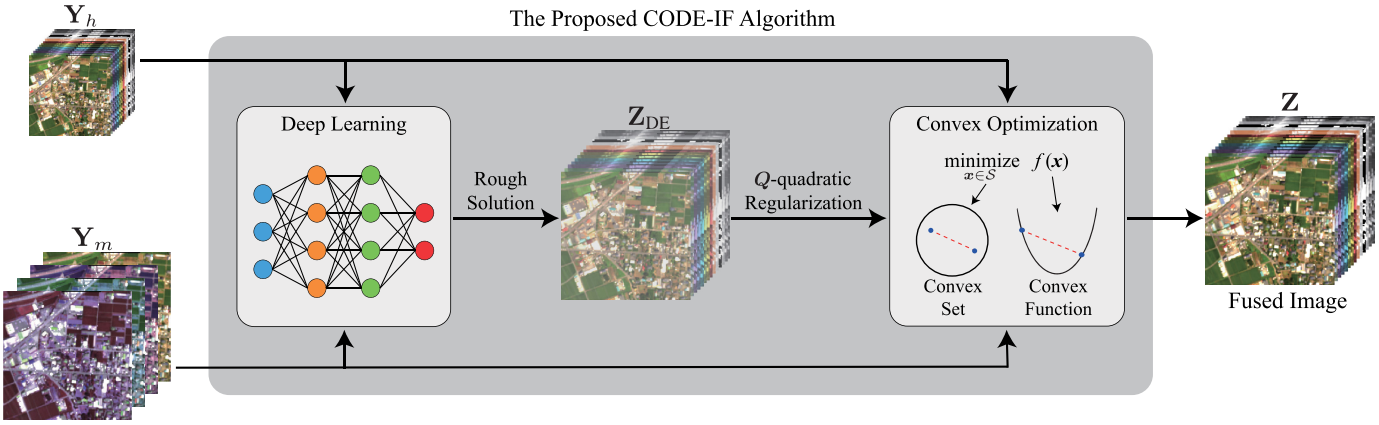


Fig. 2. Graphical illustration of the proposed CODE-IF algorithm. The LR HSI $\mathbf{Y}_h$, the HR MSI $\mathbf{Y}_m$, as well as useful features of the rough DE solution $\mathbf{Z}_{\mathrm{DE}}$ extracted by the $\boldsymbol{Q}$-quadratic function, are simultaneously fed into an efficient CO algorithm for producing the fused HR HSI $\mathbf{Z}$.

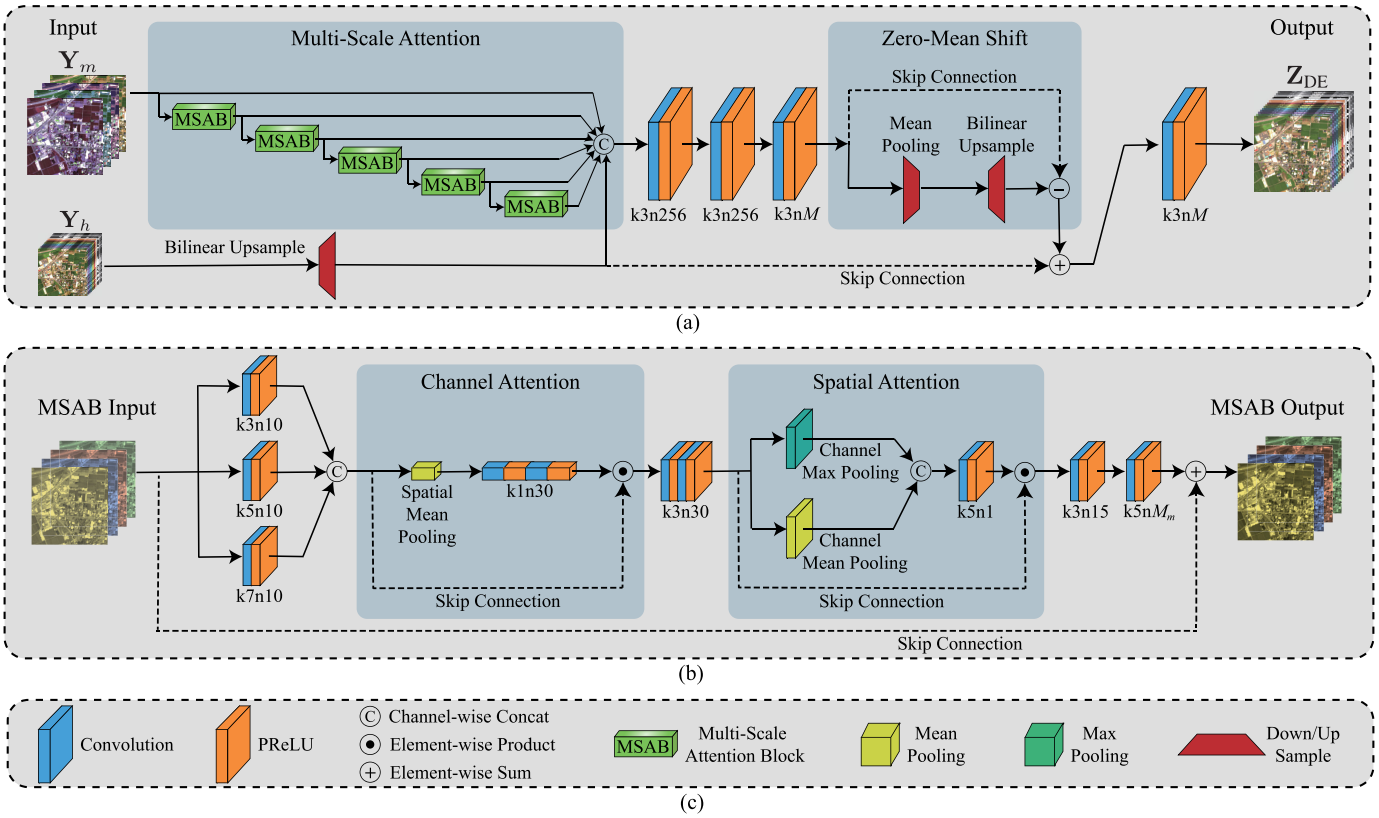


Fig. 3. Graphical illustration of the proposed deep fusion network, where "k(s)n(c)" means that the corresponding convolution layer has c output channels with a kernel size of $s \times s$, and PReLU means the well-known parametric rectified linear unit. (a) Proposed network architecture under the CODE theoretical framework. (b) Details of the multiscale attention block (MSAB). (c) Legends and annotations of the network elements.

B. Deep Regularization and Implementation Details

In this section, we explain how $\mathbf{Z}_{\mathrm{DE}}$ is computed for deep regularization, derive closed-form algorithmic solutions for solving (13), and refine the convex fusion algorithm to achieve low computational complexity implementations.

As discussed in Section II-C, under the CODE theoretical framework, a simple network architecture is sufficient for effective regularization. In this work, we design the fusion network by simply combining the multiscale learning [52], the zero-centric learning [53] and the residual learning [54], as shown in Fig. 3(a). The multiscale attention block (MSAB) and the zero-mean shift (ZMS) module are further detailed below. The MSAB architecture design is illustrated in Fig. 3(b), in which convolutions with different kernel sizes are first conducted. Different from [52], our multiscale feature maps are first processed by some attention mechanisms before they are fed into the next stage. Specifically, in Fig. 3(b), the channel attention block is used to highlight those important features and to suppress less important features, while the spatial attention block is used to highlight those important pixels and to suppress less important pixels. The channel attention unit assigns different weights to each channel of the feature map to improve its representation ability, which is achieved by element-wise multiplication with the weight obtained by pooling the map along the spatial dimension; similarly, the spatial attention map is obtained by implementing the same procedure but along the channel dimension.

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Algorithm 2 CODE-IF Algorithm for HSI Super-Resolution

1: **Given** the penalty parameter $p > 0$ and the small data solution $\mathbf{S}_{\mathrm{DE}}$. Initialize $q := 0$, $\mathbf{V}^q$, and $\mathbf{S}^q$.
2: **repeat**
3: update $\mathbf{T}^{q+1}$ by solving Eq. (15);
4: update $\mathbf{S}^{q+1}$ by solving Eq. (18);
5: update $\mathbf{V}^{q+1} := \mathbf{V}^q - (\mathbf{S}^{q+1} - \mathbf{T}^{q+1})$;
6: $q := q + 1$;
7: **until** the predefined stopping criterion is met.
8: **Output** $\mathbf{S}^\star := \mathbf{S}^q$, and the super-resolved hyperspectral image $\mathbf{Z} := \mathbf{E}\mathbf{S}^\star$.

The required network elements are collectively presented in Fig. 3(c). Regarding the ZMS module, it is used to reduce the burden of learning residual information. Different from [53], CODE-IF takes the average patch-by-patch, better filtering and enhancing the local information, which should benefit the subsequent super-resolution task. The method overlaps the learned zero-mean HR residual image with a mean-value invariant upsampled LR HSI to inject the missing high-frequency details. The ZMS process of obtaining the zero-mean residual map involves subtracting the mean value map from the HR feature map. To generate the mean value map, the HR feature map is initially downsampled to LR size through mean-pooling to determine the mean value of the HR $r \times r$ patch corresponding to each LR pixel, and then the map is bilinearly upsampled to match the size of the HR map. This simple trick not only allows the model to focus on learning high-frequency spatial details, but also enhances the network convergence speed. Finally, the output residual map is merged to the upsampled LR HSI to generate the rough solution $\mathbf{Z}_{\mathrm{DE}}$, as illustrated in Fig. 3(a). The network is trained in an end-to-end manner by minimizing the ℓ_1 loss function $\|\widetilde{\mathbf{Z}} - \widetilde{\mathbf{Z}}_{\mathrm{DE}}\|_1$, where $\widetilde{\mathbf{Z}}$ is the reference HR HSI and $\widetilde{\mathbf{Z}}_{\mathrm{DE}}$ is the network output during the training stage.

The remaining parts are to finalize the design of the CO algorithm, as well as to reduce the computational complexity. According to the theory recalled in Section II-C, one can solve (13) using Algorithm 2, where the augmented Lagrangian can be explicitly expressed as

$$\mathcal{L}(\mathbf{S}, \mathbf{T}, \mathbf{V}) \triangleq \mathrm{DF}(\mathbf{S}) + \frac{\lambda}{2}\|\mathrm{vec}(\mathbf{E}\mathbf{T} - \mathbf{Z}_{\mathrm{DE}})\|_{\boldsymbol{Q}}^2 + \frac{p}{2}\|\mathbf{S} - \mathbf{T} - \mathbf{V}\|_F^2 \tag{14}$$

where $\mathbf{V}$ is the scaled dual-variable, $\mathbf{T}$ is a newly introduced primal variable for reformulating (13) into the ADMM-form [11], the PSD matrix $\boldsymbol{Q}$ is chosen according to the original CODE theory [18, Sec. II.C] as $\boldsymbol{Q} \triangleq \mathbf{I}_L \otimes (\mathbf{E}\mathbf{E}^T)$, $\lambda > 0$ is the regularizer parameter, and $p > 0$ is the penalty parameter. We still need to derive closed-form solutions for $\mathbf{T}^{q+1}$ and $\mathbf{S}^{q+1}$ to complete Algorithm 2. First, we derive $\mathbf{T}^{q+1}$ as follows:

$$\begin{aligned}\mathbf{T}^{q+1} &:= \arg\min_{\mathbf{T}} \mathcal{L}\left(\mathbf{S}^q, \mathbf{T}, \mathbf{V}^q\right) \\ &= \arg\min_{\mathbf{T}} \lambda\|\mathrm{vec}(\mathbf{E}\mathbf{T} - \mathbf{Z}_{\mathrm{DE}})\|_{\boldsymbol{Q}}^2 + p\|\mathbf{S}^q - \mathbf{T} - \mathbf{V}^q\|_F^2\end{aligned} \tag{15}$$

whose closed-form expression can be derived using Property 2 that is proved in Appendix B.

Property 2: Given the eigenspace representation $\mathbf{Z}_{\mathrm{DE}} = \mathbf{E}\mathbf{S}_{\mathrm{DE}}$, we have the relation of

$$\|\mathrm{vec}(\mathbf{E}\mathbf{T} - \mathbf{Z}_{\mathrm{DE}})\|_{\boldsymbol{Q}}^2 = \|\mathbf{T} - \mathbf{S}_{\mathrm{DE}}\|_{\mathrm{F}}^2 \tag{16}$$

where $\boldsymbol{Q} \triangleq \mathbf{I}_L \otimes (\mathbf{E}\mathbf{E}^T)$ is a PSD matrix designed based on the CODE theory [18, Sec. II.C]. □

By Property 2, one can easily prove that $\mathbf{T}^{q+1}$ is a convex combination of $\mathbf{S}_{\mathrm{DE}}$ and $\boldsymbol{\Phi}^q \triangleq \mathbf{S}^q - \mathbf{V}^q$; to be precise, we have

$$\mathbf{T}^{q+1} := \frac{\lambda}{\lambda + p}\mathbf{S}_{\mathrm{DE}} + \frac{p}{\lambda + p}\boldsymbol{\Phi}^q. \tag{17}$$

Next, we derive $\mathbf{S}^{q+1}$, as well as its fast implementation, as follows. We first explicitly write down the corresponding optimization subproblem

$$\begin{aligned}\mathbf{S}^{q+1} &:= \arg\min_{\mathbf{S}} \mathcal{L}\left(\mathbf{S}, \mathbf{T}^{q+1}, \mathbf{V}^q\right) \\ &= \arg\min_{\mathbf{S}} \mathrm{DF}(\mathbf{S}) + \frac{p}{2}\|\mathbf{S} - \mathbf{T}^{q+1} - \mathbf{V}^q\|_F^2\end{aligned} \tag{18}$$

whose closed-form expression can be derived using Property 3 that is proved in Appendix C.

Property 3: Given the vectorized hyperspectral and multispectral data $\mathbf{y} \triangleq [\mathbf{y}_h^T,\ \mathbf{y}_m^T]^T$ with $\mathbf{y}_h \triangleq \mathrm{vec}(\mathbf{Y}_h)$ and $\mathbf{y}_m \triangleq \mathrm{vec}(\mathbf{Y}_m)$, the data-fitting term (12) can be greatly simplified as

$$\mathrm{DF}(\mathbf{s}) = \|\mathbf{y} - \mathbf{C}\mathbf{s}\|_2^2 \tag{19}$$

where $\mathbf{s} \triangleq \mathrm{vec}(\mathbf{S})$ and $\mathbf{C} \triangleq [(\mathbf{B}^T \otimes \mathbf{E})^T,\ (\mathbf{I}_L \otimes \mathbf{D}\mathbf{E})^T]^T$. □

Define $\boldsymbol{\psi}^q \triangleq \mathrm{vec}(\mathbf{T}^{q+1} + \mathbf{V}^q)$. Then, by employing Property 3, we see that deriving $\mathbf{S}^{q+1}$ is equivalent to deriving its vectorized version, that is,

$$\begin{aligned}\mathbf{s}^{q+1} &= \arg\min_{\mathbf{s}} \|\mathbf{y} - \mathbf{C}\mathbf{s}\|_2^2 + \frac{p}{2}\|\mathbf{s} - \boldsymbol{\psi}^q\|_2^2 \\ &= \arg\min_{\mathbf{s}} \mathbf{s}^T\left[\mathbf{C}^T\mathbf{C} + \frac{p}{2}\mathbf{I}_{NL}\right]\mathbf{s} - 2\left[\mathbf{C}^T\mathbf{y} + \frac{p}{2}\boldsymbol{\psi}^q\right]^T\mathbf{s} \\ &= \left[\mathbf{C}^T\mathbf{C} + \frac{p}{2}\mathbf{I}_{NL}\right]^{-1}\left[\mathbf{C}^T\mathbf{y} + \frac{p}{2}\boldsymbol{\psi}^q\right].\end{aligned} \tag{20}$$

Although (20) is already a closed-form solution, one can easily verify that its computational complexity $\mathcal{O}((NL)^2(ML_h + M_mL + NL)) = \mathcal{O}(L_h(NL)^2\xi)$ (wherein $\xi \triangleq \max\{M, M_m r^2, Nr^2\}$) is very high, preventing it from being applied in solving practical super-resolution scenarios. Therefore, we need the following lemma to implement an equivalent (but with much lower computational complexity) closed-form solution of $\mathbf{s}^{q+1}$.

Lemma 1: Let $\bar{\mathbf{C}} \triangleq [(\mathbf{g}^T \otimes \mathbf{E})^T,\ (\mathbf{I}_{r^2} \otimes \mathbf{D}\mathbf{E})^T]^T \in \mathbb{R}^{(M+r^2M_m)\times(r^2N)}$, where $\mathbf{g} \in \mathbb{R}^{r^2}$ is the vectorized blurring kernel corresponding to the blurring matrix $\mathbf{B}$ [3]. Then, (20) is equivalent to

$$\begin{aligned}\mathbf{s}^{q+1} := {} & \frac{p}{2}\left[\mathbf{I}_{L_h} \otimes \left(\bar{\mathbf{C}}^T\bar{\mathbf{C}} + \frac{p}{2}\mathbf{I}_{r^2N}\right)^{-1}\right]\boldsymbol{\psi}^q \\ & + \left\{\mathbf{I}_{L_h} \otimes \left[\left(\bar{\mathbf{C}}^T\bar{\mathbf{C}} + \frac{p}{2}\mathbf{I}_{r^2N}\right)^{-1}(\mathbf{g}^T \otimes \mathbf{E})^T\right]\right\}\mathbf{y}_h \\ & + \left\{\mathbf{I}_{L_h} \otimes \left[\left(\bar{\mathbf{C}}^T\bar{\mathbf{C}} + \frac{p}{2}\mathbf{I}_{r^2N}\right)^{-1}(\mathbf{I}_{r^2} \otimes \mathbf{D}\mathbf{E})^T\right]\right\}\mathbf{y}_m\end{aligned} \tag{21}$$

whose computational complexity is $\mathcal{O}((NL + (Nr^2)^2)\xi)$. □

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The proof of Lemma 1 is presented in Appendix D. Notice that the complexity of (20) depends on L^2, while that of (21) depends just on L; as L is the number of pixels in the HR target image (typically a very large number), Lemma 1 greatly improves the computational efficiency when implementing the closed-form solution of $\mathbf{s}^{q+1}$. The development of the CODE-IF algorithm has, therefore, been completed.

C. Related Discussions

In this section, we make some related discussions about the proposed CODE-IF. CODE-IF is particularly suitable for hyperspectral information processing because the PSD matrix $\boldsymbol{Q}$ embedded in the CO criterion, acting as a regularizer, is carefully designed according to the natural of hyperspectral data [18]. We will conduct comprehensive experimental evidence to show that significant improvements can be achieved by our CO model, even for those DE-based methods trained by big data or other fast simple fusion approaches (see Section IV-C). In other words, the DE solution used to compute $\mathbf{T}^{q+1}$ in Algorithm 2 can be replaced by the fusion results yielded by other fusion methods, thereby using Algorithm 2 to subsequently help those fusion methods to improve their fusion performances.

We remark that CODE-IF or other CODE-driven frameworks can be regarded as DE-aided CO methods. By contrast, we would like to mention that there also exist some CO-aided DE methods [55], which may also be a good way to solve remote sensing problems. However, implementing CO-aided DE is nontrivial, and it is a completely different line, which is left as our future work. Once the first CODE work was proposed for satellite data restoration in the remote sensing literature [18], it has been used to invent numerous CODE-based technologies with state-of-the-art performances [21], [56], [57]. For example, in [21], the hardware design in the metasurface-driven miniaturized hyperspectral satellite does not have related big data for the essential computational imaging, and hence CODE theory is applied there to solve such a dilemma with small-data learning. For another example, the works [56] and [57], respectively, proposed a mangrove mapping algorithm and a change detection algorithm, both are CODE-driven and have achieved state-of-the-art performances. Besides the fusion algorithm (CODE-IF) proposed in this article, we expect that the CODE theory will spur broader applications in related fields.

IV. Experimental Results

In this section, we will compare the performance of the proposed CODE-IF algorithm with other benchmark fusion methods based on Wald's protocol [58], where the experimental setting will be presented in Section IV-A. We will demonstrate that even with much less amount of training data (about 10%), CODE-IF still outperforms peer methods, as discussed in Section IV-B. In Section IV-C, we further investigate the model selection (e.g., ablation study) and model-order selection for CODE-IF. Lastly, in Section IV-D, we will apply CODE-IF to real-world scenes through real data experiments.

A. Experimental Setting

In this section, we apply Wald's protocol [58] to investigate the performance of CODE-IF. Additionally, to demonstrate the efficacy of CODE-IF across diverse types of HSIs, three benchmark remote sensing datasets and CAVE dataset [59] are regarded as reference HR HSI $\mathbf{Z}$ in our experiment, as introduced below.

1) The Chikusei data [60] was acquired by the Headwall Hyperspec-VNIR-C imaging sensor in 2014. After removing some corrupted boundaries, a 2517×2335 image with 128 spectral bands (ranging from 363 to 1018 nm) is used in the experiment. Then, we design a uniform spectrally downsampling matrix $\mathbf{D} \in \mathbb{R}^{4\times 128}$ corresponding to the Landsat TM bands 1–4 (covering 450–520, 520–600, 630–690, and 760–900 nm regions, respectively) to generate HR MSI $\mathbf{Y}_m$. The 1200 overlapped 64×64 patches are cropped from the lower 1240×2200 area for training, and one 200×200 patch from the remaining area is selected for testing. All the 1200 data are used to train the pure DE-based peer methods (i.e., D-UNet, SSR-Net, MSDCNN, ResTFNet, MHFnet, and HSRnet), but, among which, only 150 data (i.e., 12.5%) are used to train CODE-IF.
2) The Washington DC Mall data [61] was obtained by hyperspectral digital imagery collection experiment (HYDICE) sensor in 1995, from which a 1280×307 image with 191 spectral bands (ranging from 400 to 2500 nm) is used in the experiment. A uniform spectrally downsampling matrix $\mathbf{D} \in \mathbb{R}^{6\times 191}$ is designed corresponding to the Landsat TM bands 1–5 and 7 (covering 450–520, 520–600, 630–690, 760–900, 1550–1750, and 2080–2350 nm regions, respectively) to generate $\mathbf{Y}_m$. Similarly, the training patches are cropped from the lower 1050×307 area, while the other settings are the same as in the Chikusei dataset.
3) The 2013 IEEE Geoscience and Remote Sensing Symposium (GRSS) Data Fusion Contest Dataset at the University of Houston (UH) [62] was taken by the NSF-funded Center for Airborne Laser Mapping (NCALM) in June 2012. After removing corrupted boundaries, a 349×1905 image with 144 bands (ranging from 380 to 1050 nm) is used in the experiment. Then, we design a uniform spectrally downsampling matrix $\mathbf{D} \in \mathbb{R}^{4\times 144}$ corresponding to the Landsat TM bands 1–4 (covering 450–520, 520–600, 630–690, and 760–900 nm regions, respectively) to generate $\mathbf{Y}_m$. The training patches are cropped from the right 349×1205 area, while the other settings are the same as in the Chikusei dataset.
4) The CAVE dataset [59] was captured by Apogee Alta U260 in 2008, from which a total of 32 HSIs with 512×512 pixels and 31 bands (ranging from 400 to 700 nm) was used. Then, we design a spectrally downsampling matrix $\mathbf{D} \in \mathbb{R}^{3\times 31}$ based on the response of the Nikon D700 camera to generate $\mathbf{Y}_m$. The 1280 nonoverlapped 64×64 training patches are

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TABLE I
DIMENSION OF THE FOUR SIMULATION DATA

Dataset	$\mathbf{Z}$	$\mathbf{Y}_h$	$\mathbf{Y}_m$
Chikusei	200×200×128	50×50×128	200×200×4
Washington DC Mall	200×200×191	50×50×191	200×200×6
Houston University	200×200×144	50×50×144	200×200×4
CAVE	512×512×31	128×128×31	512×512×3

cropped from 20 HSIs (64 patches per image) [31]. To train the proposed small-data-driven network, only 192 patches (i.e., 15%) cropped from only three HSIs (i.e., *oil painting, glass tiles*, and *fake and real strawberries*) are used, while the HSI "*fake and real lemon slices*" is for testing.

In each dataset, the LR HSI $\mathbf{Y}_h$ is obtained by performing spatial blurring with a Gaussian blur filter (variance 2 and blurring factor 4) on $\mathbf{Z}$, and the details of all related data dimensions are summarized in Table I. Although we have noticed other filters like box and Starck and Murtagh [63] ones, the most commonly used one is the Gaussian one like in the seminal works [16], [45], [64]. Real-data experiments will also support our model selection of the Gaussian blur filter (see Section IV-D). In addition, a zero-mean additive Gaussian noise with an SNR 40 dB is added to both $\mathbf{Y}_m$ and $\mathbf{Y}_h$, respectively [65], to capture the modeling error in (6)–(7).

All the experiments are conducted under MATLAB R2021b, PyTorch 3.8, and Tensorflow 2.8.0 on a personal computer equipped with an Intel i9-10900K CPU, 64 GB system memory, and NVIDIA RTX-3090 GPU. For a fair comparison, all the pure DE-based methods are trained with batch size 15 for a total of 48 000 iterations or until they converge. Also, all the training is done by ADAM optimizer with the learning rate reduced according to a cosine scheduler from 10^{-4} to 10^{-6}. Data augmentation methods such as random horizontal/vertical flips and rotation are applied when training.

Four frequently used evaluation metrics are adopted to quantitatively evaluate the fusion results, based on the similarity between the reference image $\mathbf{Z}$ and its estimate $\hat{Z}$. Their definitions are listed below for rigorousness.

1) The peak signal-to-noise ratio (PSNR) is defined as $\text{PSNR} = (1/M)\sum_{m=1}^{M}\text{PSNR}_m$, where PSNR_m measures the spatial quality of the mth band, that is,

$$\text{PSNR}_m = 10\log_{10}\left(\frac{\max\left\{\left(\hat{z}_{mn}\right)^2 \mid n \in \mathcal{I}_L\right\}}{\frac{1}{L}\|\hat{Z}^{(m)} - \mathbf{Z}^{(m)}\|_2^2}\right)$$

where $\hat{z}_{mn}$ denotes the nth entry in the vector $\hat{Z}^{(m)}$, and $\mathcal{I}_L \triangleq \{1, \ldots, L\}$. A higher PSNR value indicates a better spatial quality of the fused image $\hat{Z}$.

2) The spectral angle mapper (SAM) is defined as

$$\text{SAM} = \frac{1}{L}\sum_{n=1}^{L}\arccos\left(\frac{\left(\hat{z}[n]\right)^T(\mathbf{z}[n])}{\|\hat{z}[n]\|_2 \cdot \|\mathbf{z}[n]\|_2}\right)$$

where $\hat{z}[n]$ denotes the nth column of $\hat{Z}$. A smaller value of SAM indicates better spectral shape preservation of the fused result $\hat{Z}$.

3) The root mean squared error (RMSE) is defined as $\text{RMSE} = ((1/M)\sum_{m=1}^{M}\text{RMSE}_m^2)^{1/2}$, where the error of the mth band is given by $\text{RMSE}_m = (1/(L)^{1/2})\|\hat{Z}^{(m)} - \mathbf{Z}^{(m)}\|_2$. A smaller RMSE value means better global fusion quality.

4) A global quality measure, called *erreur relative globale adimensionnelle de synthèse* (ERGAS), is defined as

$$\text{ERGAS} = \frac{100}{r}\sqrt{\frac{1}{M}\sum_{m=1}^{M}\frac{\text{RMSE}_m^2}{\mu_{\mathbf{Z}^{(m)}}^2}}$$

where r is the resolution ratio between LR and HR counterparts and $\mu_{\mathbf{Z}^{(m)}}^2$ is the mean of $\mathbf{Z}^{(m)}$. The smaller the value of ERGAS, the lower the degree of global distortion in the fused image $\hat{Z}$.

B. Quantitative and Qualitative Performance Assessments

In this section, the proposed CODE-IF is compared with several state-of-the-art fusion methods for subjective and objective quality assessments. The peer methods include D-UNet [17], SSR-Net [29], MSDCNN [14], ResTFNet [15], MHFnet [31], HSRnet [16], CNN-Fus [12], HySure [8], R-FUSE [64], [66], CO-CNMF [3], GSA [67], and LTMR [22]. As reported in Section IV-A, the proposed CODE-IF is trained using only small data, which is about 10% of the training data used to train other DE-based methods. Moreover, due to our small-data-driven simple network architecture, the average training time and the testing time over four datasets are 7271.5720 s (just about 2 h) and 0.0064 s, respectively, which means that training CODE-IF does not need to consume a lot of computational resources. This shows another great advantage (of the CODE theory) that training a small-data-driven network architecture is indeed very fast, and this is critical once we have some commercial purposes like speeding up software production. Even with such a limited amount of training data, CODE-IF still yields superior fusion performance, as will be seen below.

The fusion results are visually displayed as the false-color composition of the super-resolved HSI $\mathbf{Z}$ for the Chikusei data (see Fig. 4), the Washington DC Mall data (see Fig. 5), the Houston University data (see Fig. 6), and the CAVE data (see Fig. 7). The tested region of interest (ROI) for each data has been displayed in the top-left subfigure (see Figs. 4–7). As one can see, all the fusion methods under test yield very pleasant qualitative results for all four datasets. However, the good results are not distinguishable by just visualizing the false-color composition. Thus, we adopt another intuitive and probably more easily understandable visualization of the residual map, whose nth pixel is defined as the so-called relative absolute error (RAE) $R_n \triangleq (\sum_{b=1}^{M}(\hat{Z}_{b,n} - \mathbf{Z}_{b,n})^2)^{1/2}/(\sum_{b=1}^{M}(\mathbf{Z}_{b,n})^2)^{1/2}$, with b being the band index. So, in each subfigure of Figs. 4–7, the upper half is the false-color composition of the fusion result, while the lower half is the fusion residual error map with color bar shown in the right part of each figure. From the residual map, one can clearly see that CODE-IF indeed more accurately reconstructs the spatial details, as can be seen from the lower-right subfigures in Figs. 4–7.

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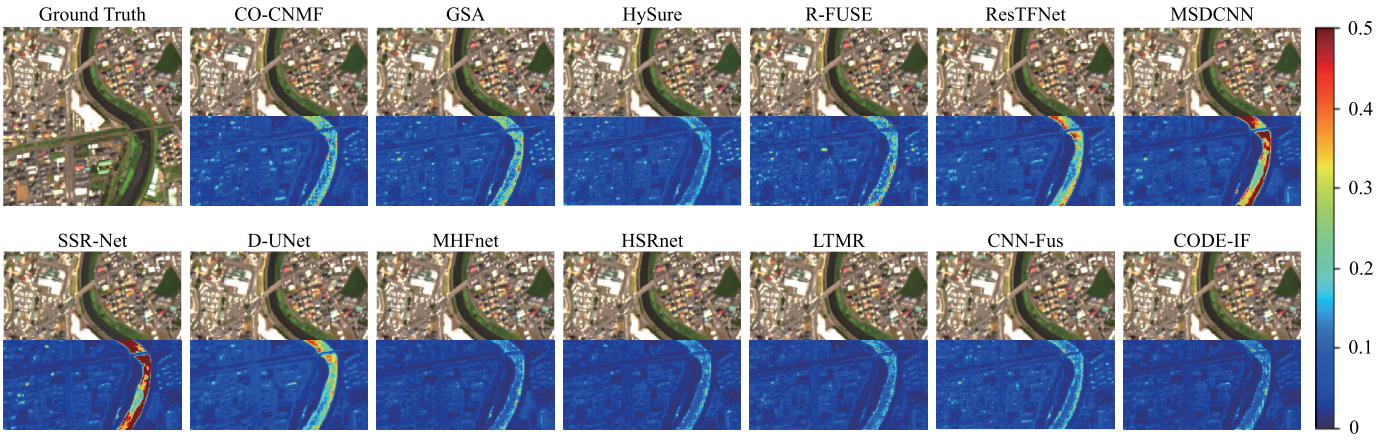


Fig. 4. Fusion results [false-color composition: B55(r)-B35(g)-B11(b)] and residual maps for the Chikusei data, where CODE-IF uses just 150 data, while the other DE-based methods all use 1200 data.

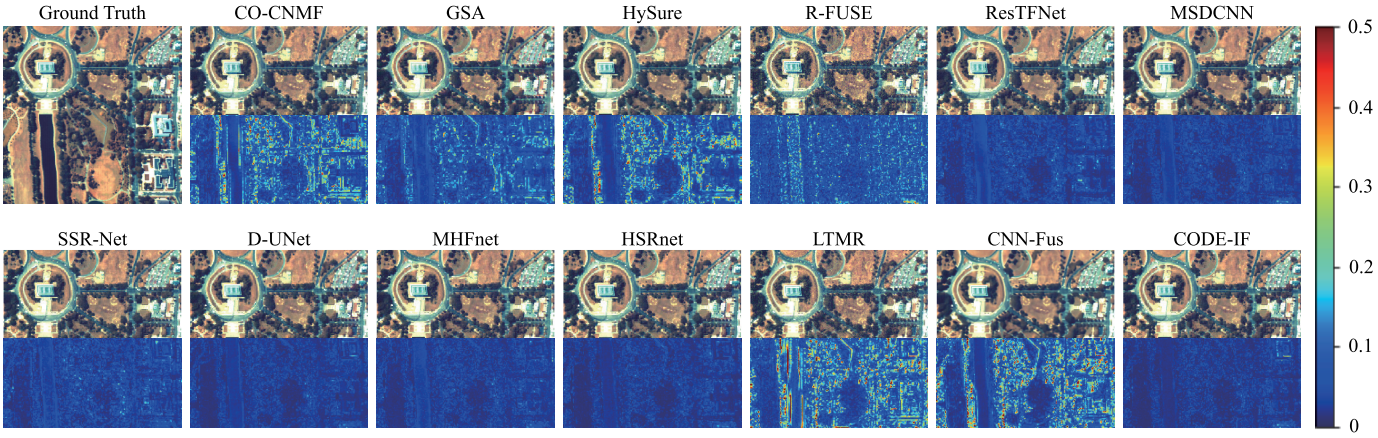


Fig. 5. Fusion results [false-color composition: B55(r)-B35(g)-B11(b)] and residual maps for the Washington DC Mall data, where CODE-IF uses just 150 data, while the other DE-based methods all use 1200 data.

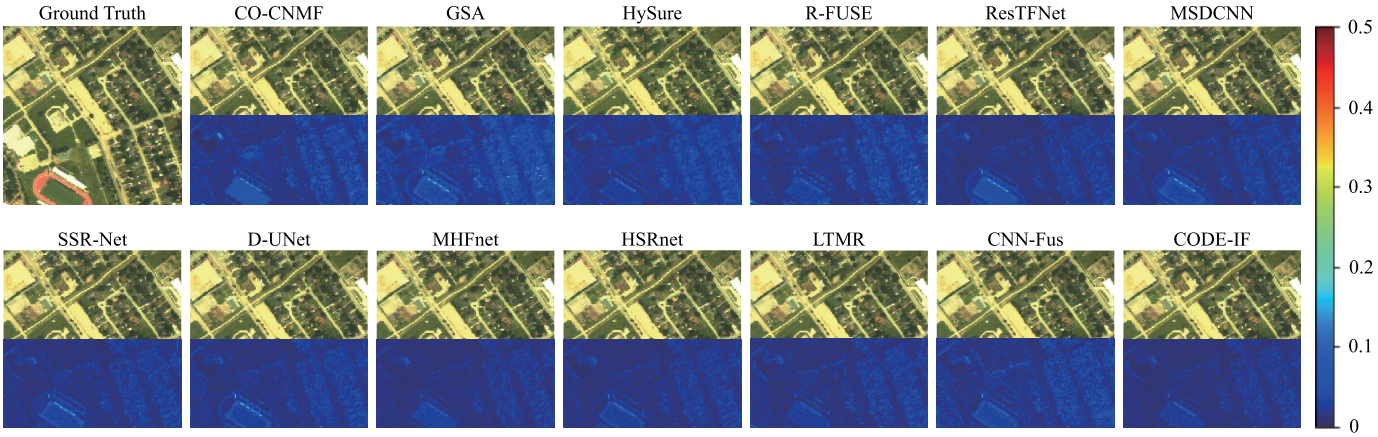


Fig. 6. Fusion results [false-color composition: B55(r)-B35(g)-B11(b)] and residual maps for the Houston University data, where CODE-IF uses just 150 data, while the other DE-based methods all use 1200 data.

To fairly argue that CODE-IF does outperform benchmark HSI/MSI fusion methods, we also need to investigate the quantitative evaluation. So, the testing results in terms of PSNR, SAM, RMSE, and ERGAS are summarized in Tables II (Chikusei data), III (Washington DC Mall data), IV (Houston University data), and V (CAVE data). The red color number means the best performance of some algorithm, while the blue color number means that the algorithm performs second in a specific scenario. One can see that CODE-IF yields excellent spatial quality as indicated by the highest or nearly highest PSNR values in Tables II, IV, and V. CODE-IF also performs favorably in fusing the spectral information as indicated by the lowest or nearly lowest SAM values in Tables II–V. In terms of RMSE and EGRAS, one can see

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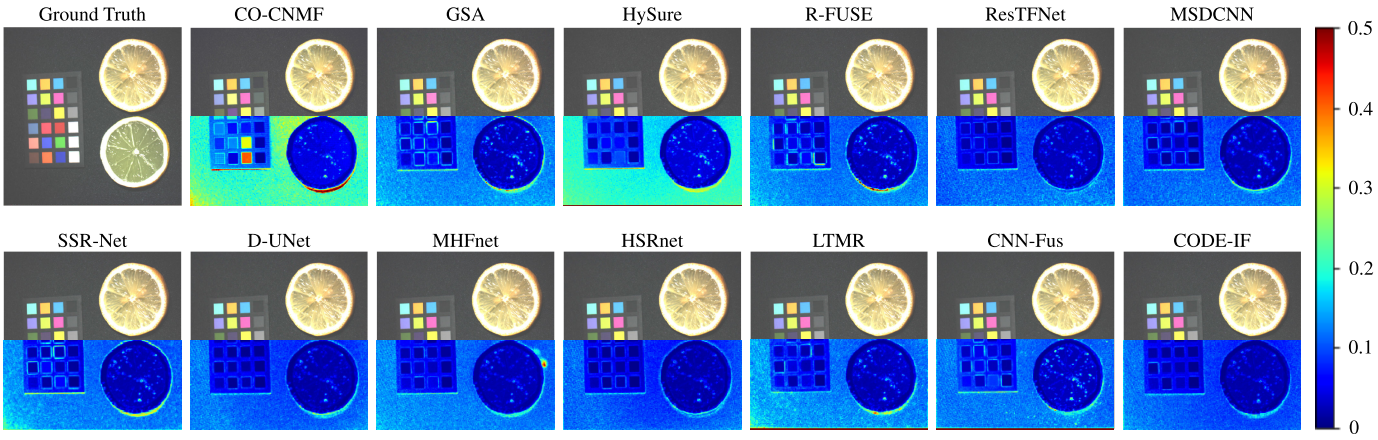


Fig. 7. Fusion results [false-color composition: B22(r)-B14(g)-B7(b)] and residual maps for the CAVE data, where CODE-IF uses just 192 data, while the other DE-based methods all use 1280 data.

TABLE II

PERFORMANCE COMPARISON OF DIFFERENT METHODS ON CHIKUSEI DATA; CODE-IF IS TRAINED USING JUST 150 DATA, WHILE THE OTHER SIX PURE DE-BASED METHODS ALL USE 1200 TRAINING DATA

Methods	Performance Measures				Time (sec.)
	PSNR	SAM	RMSE	ERGAS	
D-UNet	45.1698	2.8358	54.0348	1.8532	0.007
SSR-Net	46.2864	2.9733	52.3738	1.8663	0.001
MSDCNN	45.7106	3.2043	54.4414	1.9893	0.002
ResTFNet	44.9944	2.7712	58.0407	2.2101	0.003
MHFnet	47.7069	2.0260	46.9783	1.8879	0.551
HSRnet	48.7811	1.9567	42.0139	1.6062	0.223
CNN-Fus	45.9072	2.4340	54.9709	2.0833	259.394
HySure	45.9363	2.6927	55.5424	2.1657	10.786
R-FUSE	45.9652	2.7464	60.5593	2.1660	0.744
CO-CNMF	45.3767	2.9374	56.0428	2.1044	3.733
GSA	45.9411	2.9412	59.8525	1.8798	0.295
LTMR	48.5630	2.1071	43.8550	1.7482	36.112
CODE-IF	49.2484	1.8257	40.0761	1.5398	0.252

TABLE III

PERFORMANCE COMPARISON OF DIFFERENT METHODS ON WASHINGTON DC MALL DATA; CODE-IF IS TRAINED USING JUST 150 DATA, WHILE THE OTHER SIX PURE DE-BASED METHODS ALL USE 1200 TRAINING DATA

Methods	Performance Measures				Time (sec.)
	PSNR	SAM	RMSE	ERGAS	
D-UNet	41.7374	1.3460	94.1541	1.3513	0.008
SSR-Net	37.2849	1.7604	139.0431	5.2630	0.001
MSDCNN	37.1739	1.5934	115.8459	3.2152	0.002
ResTFNet	37.3978	1.6588	128.9336	2.6769	0.003
MHFnet	37.6249	1.4667	104.7261	5.8954	0.640
HSRnet	42.0632	1.2823	96.5697	0.9883	0.245
CNN-Fus	34.1229	4.0630	336.6730	3.2074	275.821
HySure	33.5855	3.3077	264.4024	3.0417	10.143
R-FUSE	36.6992	3.4783	256.7665	2.2240	0.771
CO-CNMF	33.0333	3.7772	297.8747	3.2741	4.015
GSA	38.7277	2.8273	224.4854	1.7300	0.554
LTMR	35.5289	4.4852	332.5251	2.7171	36.176
CODE-IF	41.0850	1.2559	94.7032	1.2439	0.287

TABLE IV

PERFORMANCE COMPARISON OF DIFFERENT METHODS ON HOUSTON DATA; CODE-IF IS TRAINED USING JUST 150 DATA, WHILE THE OTHER SIX PURE DE-BASED METHODS ALL USE 1200 TRAINING DATA

Methods	Performance Measures				Time (sec.)
	PSNR	SAM	RMSE	ERGAS	
D-UNet	46.4333	1.1818	155.5301	0.7589	0.008
SSR-Net	46.7994	1.1227	153.3008	0.7840	0.001
MSDCNN	46.3302	1.1791	159.4136	0.8120	0.002
ResTFNet	46.3418	1.1165	153.0324	0.8073	0.003
MHFnet	48.6280	0.9548	125.7743	0.6716	0.414
HSRnet	47.6453	1.0274	138.0975	0.6741	0.240
CNN-Fus	44.9888	1.3248	181.7318	1.0923	256.906
HySure	44.1032	1.4748	191.6230	1.2345	10.881
R-FUSE	44.7702	1.3520	194.0564	1.0792	0.723
CO-CNMF	45.0182	1.4717	182.7237	1.2927	3.953
GSA	44.3724	1.8337	232.5318	1.0100	0.328
LTMR	47.2342	1.1196	151.9197	0.8682	33.841
CODE-IF	48.8746	0.9225	125.6804	0.6580	0.262

TABLE V

PERFORMANCE COMPARISON OF DIFFERENT METHODS ON CAVE DATA; CODE-IF IS TRAINED USING JUST 192 DATA, WHILE THE OTHER SIX PURE DE-BASED METHODS ALL USE 1280 TRAINING DATA

Methods	Performance Measures				Time (sec.)
	PSNR	SAM	RMSE	ERGAS	
D-UNet	49.5452	4.6761	235.0454	1.4852	0.008
SSR-Net	47.4770	5.4461	304.4953	1.8948	0.001
MSDCNN	48.2627	4.7607	269.8820	1.7045	0.002
ResTFNet	47.6701	4.6677	290.9616	1.8456	0.003
MHFnet	48.8319	5.2225	256.1563	1.5719	0.840
HSRnet	49.5783	4.3130	223.6659	1.3977	0.523
CNN-Fus	46.7265	5.4353	370.3468	2.3091	820.384
HySure	46.0683	8.1416	390.5502	2.4785	86.692
R-FUSE	46.6933	5.6509	353.3647	2.2279	1.890
CO-CNMF	43.8264	8.5930	496.5769	3.0005	97.308
GSA	45.0877	5.8675	385.3476	2.4142	0.604
LTMR	47.4367	5.8165	313.6242	1.9426	154.915
CODE-IF	50.0259	4.4236	223.4520	1.3954	0.810

that CODE-IF has strong global reconstruction results as well. Remarkably, HSRnet slightly outperforms CODE-IF in the Washington DC Mall data (see Table III), while CODE-IF outperforms it in the remaining data, particularly in Chikusei data (e.g., all the quantitative indices are significantly better in Table II). We would like to draw the reader's attention that CODE-IF is trained using only about 10% amount of data

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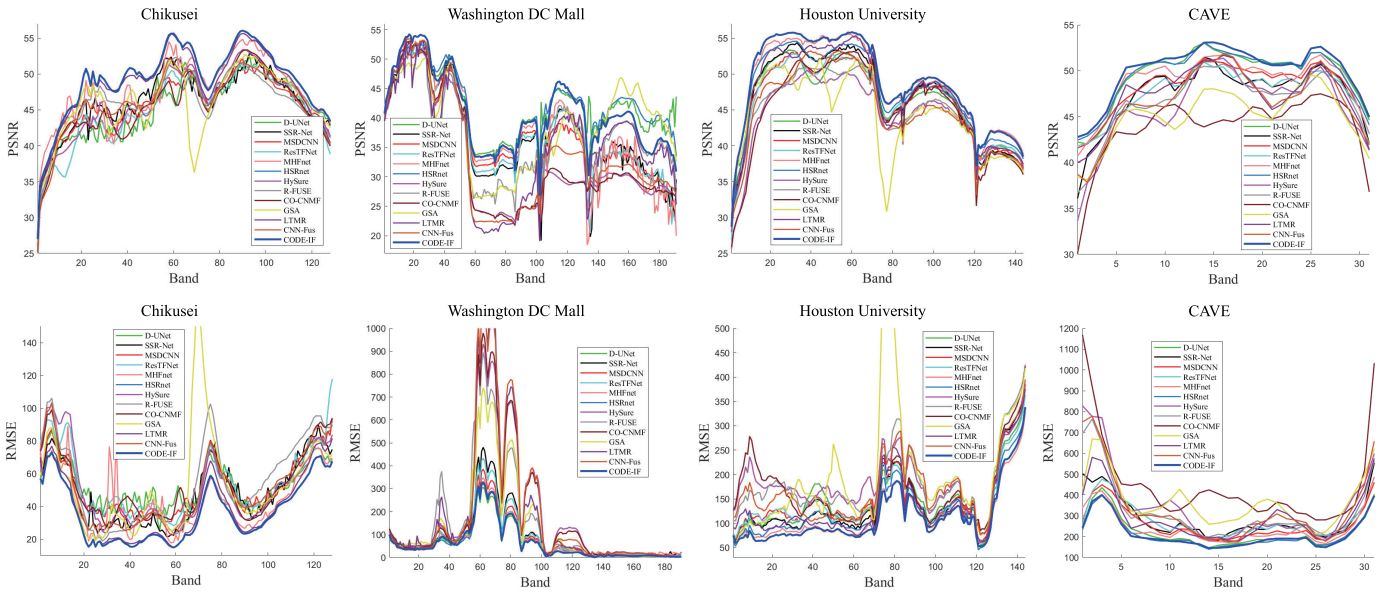


Fig. 8. Band-by-band analysis in terms of PSNR_m and RMSE_m for different fusion methods, including the proposed CODE-IF (trained on 150 or 192 data) and six pure DE-based methods (all trained on 1200 or 1280 data).

when compared to other DE-based methods (i.e., D-UNet, SSR-Net, MSDCNN, ResTFNet, MHFnet, HSRnet, and CNN-Fus). Even so, CODE-IF still slightly outperforms HSRnet, or even performs clearly better than other DE-based methods, as can be seen from Table V.

Since PSNR and RMSE can be defined band-by-band (see Section IV-A), their values are plotted as a function of the band index in Fig. 8, to have a deeper understanding of these fusion methods. Fig. 8 indicates that for HSI bands that are not overlapped with MSIs (such as bands 30–36, 51–96, 117–149, and 178–191 in the Washington DC Mall data), traditional methods may have higher distortion than DE-based methods [14]. The proposed CODE-IF algorithm, which simultaneously blends traditional theory and learning-based theory, effectively addresses the above issue and has leading fusion performance in most spectral bands.

Since CO-based methods usually involve iterative algorithmic procedures, the computational speed is often slower than pure DE-based ones. As CODE-IF blends both CO and DE theories, the computational speed is in between them. However, CODE-IF adopts a regularization scheme of a very simple mathematical form. So, CODE-IF is much faster than those non-DE-based methods (i.e., HySure, R-FUSE, CO-CNMF, GSA, and LTMR), even on par with some DE-based ones (i.e., MHFnet, HSRnet, and CNN-Fus), as can be seen from Tables II–V.

C. Model-Order Selection and Ablation Studies

In the DE part of CODE-IF, as carefully discussed in Section II-C, a naive DE network is sufficient for effective regularization under the CODE learning theory. Though the network architecture is not the key for a CODE-based algorithm, we show that the designed network is already a very concise one. Specifically, we show that the two core modules in Fig. 3 (i.e., MSAB and ZMS) are both necessary. Based on whether each module is on or off, the four combinations tested on the Chikusei dataset yield different fusion performances, as presented in Table VI. One can see that when both the MSAB and ZMS modules are on, the fusion results are best in terms of all four performance indices. This confirms that the current DE architecture has been a compact one and has no need to be further simplified. In addition, we also experimentally decide the optimal model order (i.e., the number of MSABs), and the resultant optimal model order is found to be 5 based on the performance evaluation summarized in Table VII.

On the other hand, in the CO part of CODE-IF, three parameters need to be properly set, including the model order N, the regularization strength λ, and the penalty parameter p. We test the proposed CODE-IF algorithm across a wide range of N, λ, and p to evaluate the quality of $\mathbf{Z}$ in terms of PSNR

TABLE VI

ABLATION STUDY OF THE PROPOSED CODE-IF ALGORITHM, WHERE THE MARKER "✔" MEANS THAT THE MODULE IS USED

MSAB	ZMS	PSNR	SAM	RMSE	ERGAS
		47.5260	2.1173	48.4579	1.7136
	✔	49.0504	1.8766	41.2816	1.5626
✔		47.7428	2.0786	47.6506	1.7297
✔	✔	49.2484	1.8257	40.0761	1.5398

TABLE VII

QUANTITATIVE ASSESSMENT TO ANALYZE THE EFFECTIVENESS OF DIFFERENT MSAB NUMBERS

Performance Measures	Number of MSABs				
	1	3	5	7	9
PSNR	49.1086	49.1490	49.2484	49.1929	49.2090
SAM	1.8611	1.8532	1.8257	1.8469	1.8408
RMSE	40.9367	40.6636	40.0761	40.2424	40.3462
ERGAS	1.5672	1.5578	1.5398	1.5345	1.5132

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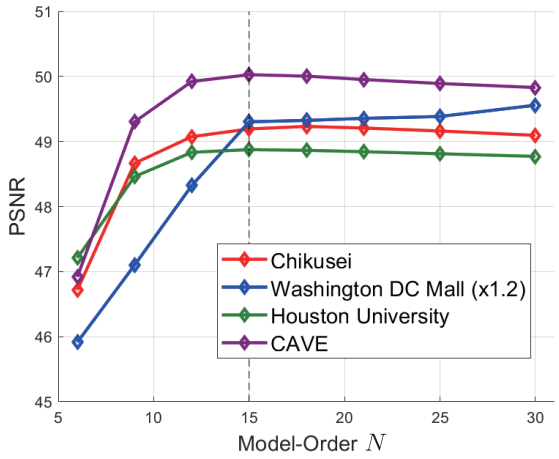


Fig. 9. Studying the model-order selection in terms of PSNR to optimize the quality of the reconstructed spatial in the fused HSI $\mathbf{Z}$.

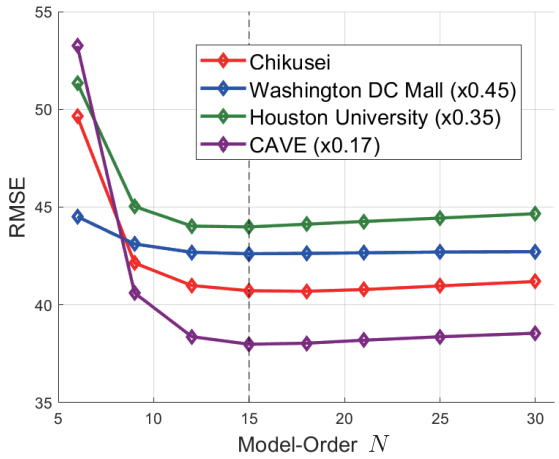


Fig. 10. Studying the model-order selection in terms of RMSE to optimize the global quality of the fused HSI $\mathbf{Z}$.

and RMSE. The experiments are conducted by varying each parameter individually while keeping the other parameters fixed. The computational complexity will become higher when N is too large, while the algorithm will not be able to well reconstruct the target HSI $\mathbf{Z}$ when N is too small. The results are summarized in Figs. 9 and 10. One can see that when $N < 15$, the reconstruction quality is not good. However, when setting N to be larger than 15, it does not help much in further improving the fusion performance. The parameter λ plays a crucial role in addressing the ill-posed problem within the framework as it determines a tradeoff between the observation model and the network solution of the final output. According to Table VIII, setting $\lambda = 0.05$ is considered the optimal choice among the four datasets. The last parameter p is a penalty parameter in the ADMM algorithm that influences the convergence speed, but as long as $p > 0$, CODE-IF must converge [11]. As the impact of p on the final performance is not as significant as that of N and λ, as shown in Table IX, we set $p = 0.1$ that yields generally good performance. Thus, we suggest future users of CODE-IF use the default settings of $N = 15$, $\lambda = 0.05$, and $p = 0.1$, which have been used throughout the experiments of this article.

TABLE VIII

PERFORMANCE OF THE PROPOSED METHOD, WITH RESPECT TO DIFFERENT PARAMETERS λ

Dataset	Performance Measures	Regularizer Parameter λ (N=15, p = 0.1)			
		0.001	0.01	0.1	1
Chikusei	PSNR	49.3098	49.3503	48.9657	47.5616
	RMSE	39.7493	39.7018	40.8331	44.9848
Washington DC Mall	PSNR	40.6381	40.9335	41.0400	40.6162
	RMSE	98.7446	96.2039	94.8164	97.0407
Houston University	PSNR	48.6568	48.7378	48.8804	48.5300
	RMSE	127.2629	126.6375	125.7823	129.3805
CAVE	PSNR	49.8490	49.9118	50.0271	49.6515
	RMSE	226.5087	225.3763	223.6748	232.3759

TABLE IX

PERFORMANCE OF THE PROPOSED METHOD, WITH RESPECT TO DIFFERENT PARAMETERS p

Dataset	Performance Measures	Penalty Parameter p (N=15, λ = 0.05)			
		0.001	0.01	0.1	1
Chikusei	PSNR	49.3572	49.2577	49.2484	49.1615
	RMSE	39.7036	39.7018	40.0516	40.3293
Washington DC Mall	PSNR	40.9337	41.0832	41.0850	41.0441
	RMSE	95.1714	94.6997	94.7032	94.6486
Houston University	PSNR	48.7631	48.8732	48.8746	48.8909
	RMSE	126.4490	125.6861	125.6804	125.6016
CAVE	PSNR	49.9310	50.0246	50.0259	50.0388
	RMSE	225.0377	223.4709	223.4520	223.3173

To demonstrate the indispensable role of the CO aspect in CODE-IF, we not only employ the proposed network solution $\mathbf{Z}_{\text{DE}}$, but also assess the performance with other network solutions produced by alternative DE-based methods on the Chikusei dataset. Apart from regarding DE-based methods as intermediate results, we also test the performance under two fast and simple models, that is, R-FUSE and GSA, to see what would happen when the CODE-IF framework does not have the DE part. Also, a comparison is made between the $\boldsymbol{Q}$-quadratic norm regularizer and the widely used TV regularizer. To be rigorous, the TV regularizer can be plugged into (1) to yield the following fusion criterion:

$$\mathbf{S}^{\star} = \arg \min_{\mathbf{z}=(\mathbf{I}_L \otimes \mathbf{E})\mathbf{s}} \text{DF}(\mathbf{s}) + \lambda \text{TV}(\mathbf{z}) \tag{22}$$

where $\text{DF}(\mathbf{s})$ is the data-fitting term defined in (12) that can be simplified as a single term via Property 3; for conciseness, the details are not presented here again. We do not show the whole derivation due to space limitations. Nevertheless, the key to solving the above TV-based fusion criterion lies within how to solve the proximal operator induced by the TV regularization, but this is a well-known operator that can be effectively implemented using the split Bregman method with available open-source code.[1] These analyses are organized in Table X, where "w/o" represents the output of each DE-based method (namely, "without CO"), "with" indicates the result obtained after using the CODE-IF framework (namely, "with CO"), and "with (TV)" denotes the result obtained using CODE-IF but replacing $\boldsymbol{Q}$-norm with TV. Results summarized

[1]This work solves TV regularization criterion by MATLAB Central File Exchange: https://www.mathworks.com/matlabcentral/fileexchange/36278-split-bregman-method-for-total-variation-denoising

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TABLE X

PERFORMANCE OF THE PROPOSED NETWORK, WITH $\mathbf{Z}_{DE}$ GENERATED BY DIFFERENT DE-BASED/NON-DE-BASED METHODS

Methods	with or w/o CODE-IF	Performance Measures			
		PSNR	SAM	RMSE	ERGAS
D-UNet	w/o	45.1698	2.8356	54.0348	1.8532
	with	48.8920	1.9644	42.7063	1.6122
SSR-Net	w/o	46.2864	2.9733	52.3738	1.8663
	with	49.1027	2.0813	41.5567	1.5733
MSDCNN	w/o	45.7106	3.2043	54.4414	1.9893
	with	48.8939	2.0633	42.6565	1.6073
ResTFNet	w/o	44.9944	2.7712	58.0407	2.2101
	with	48.4409	2.0533	44.1238	1.6722
MHFnet	w/o	47.7069	2.0260	46.9783	1.8879
	with	49.5132	1.8875	39.9338	1.5626
HSRnet	w/o	48.7811	1.9567	42.0139	1.6062
	with	49.7441	1.8200	39.0188	1.5297
R-FUSE	w/o	45.9652	2.7464	60.5593	2.1660
	with	48.1437	2.4004	49.0886	1.9065
GSA	w/o	45.9411	2.9412	59.8525	1.8798
	with	47.7184	2.2539	48.6698	1.7372
Proposed Network	w/o	46.3596	2.3316	50.3315	1.8324
	with	49.2484	1.8257	40.0761	1.5398
	with (TV)	36.7293	7.5829	230.7792	5.5139

in Table X indicate substantial performance enhancement for all DE-based/non-DE-based methods after their results are fed into our CO process. However, the performance of the final results initialized by intermediate non-DE-based solutions still does not exceed the DE ones. This means that although the CODE-IF framework has a better performance than DE-based methods, DE-aided regularization is still indispensable in such a framework. Also, the usage of $\boldsymbol{Q}$-quadratic norm is indeed significantly better than the conventional TV, just like the reason we discussed in Section II-C. Furthermore, the computational time required for TV exceeds 1000 s, rendering it impractical for real-world applications.

D. Real-Data Eexperimental Results

To further assess the performance of CODE-IF, we evaluate the real-hyperspectral data released by the 2018 IEEE GRSS Data Fusion Contest [68]. The HSI, captured by ITRES CASI 1500 imager, covers a 380–1050-nm spectral range with 48 bands at a 1-m ground sample distance (GSD). The MSI, obtained by DiMAC ULTRALIGHT+ camera, features 5-cm GSD but offers only the RGB band. Due to a lack of ground truth, we utilize the simulated Houston University dataset (see Section IV-A) for training CODE-IF, and the simulated HR HSI $\boldsymbol{Z}$ is downsampled from 144 to 48 bands to accommodate the real HSI. The generation process of the simulated HSI/MSI pair follows those described in Section IV-A, except for the blurring factor that is set to 20 (i.e., 1-m divided by 5-cm). The sizes of the two images are $4 \times 4 \times 48$ and $80 \times 80 \times 3$, respectively. Besides, we retrain our DE network to match the real HSI/MSI pair only by modifying the relevant channel number and blurring factor to match the input and output size. The 150 overlapped 80×80 training patches are cropped for retraining, and two real HSI/MSI pairs are selected for testing (see Fig. 11), in which the HSI is of size $20 \times 20 \times 48$ while the MSI is of size $400 \times 400 \times 3$.

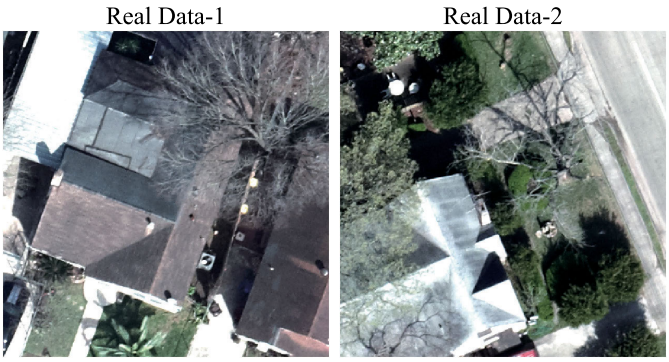


Fig. 11. True color composition of real-hyperspectral satellite data acquired over the UH and its neighboring region.

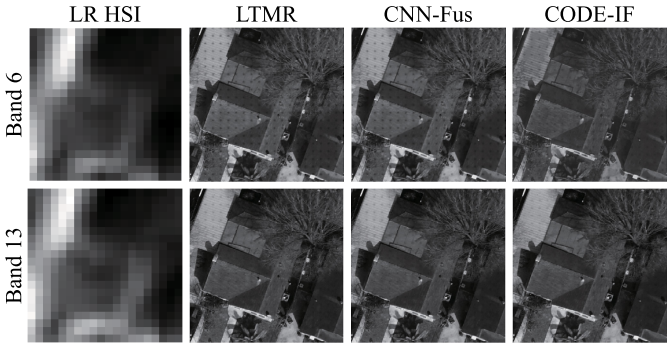


Fig. 12. First and the second rows show the qualitative results at the sixth and 13th bands for real data-1, respectively.

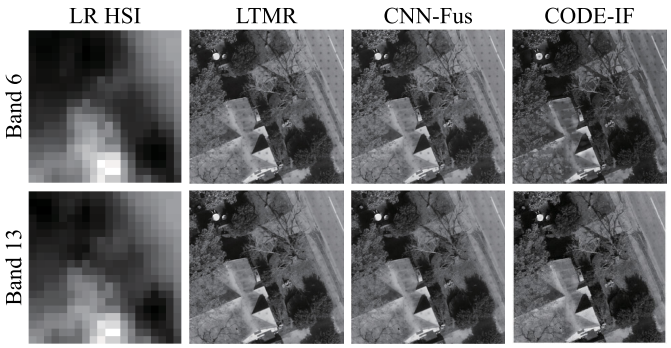


Fig. 13. First and the second rows show the qualitative results at the sixth and 13th bands for real data-2, respectively.

TABLE XI

COMPUTATIONAL TIME (IN SECONDS) OF THE TESTED METHODS

Methods	LTMR	CNN-Fus	CODE-IF
Real Data-1	87.1520	632.2770	35.7359
Real Data-2	86.8190	640.3460	35.5641

Due to the space constraints, we compare CODE-IF with one recent method from each category, including LTMR from the CO category [22] and CNN-Fus from the DE category [12]. The recovered HR HSIs are visually presented in Figs. 12 and 13, respectively, containing the sixth and 13th bands. As can be seen, all the methods can well recover the spatial details from two LR HSIs. However, due to the very large

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gap in spatial resolution (i.e., $r := 20$), we observe that regular and periodic dot artifacts slightly appear in all methods. In Fig. 12, CODE-IF yields significantly better visual quality than the other peer methods. Moreover, Table XI highlights the significantly higher computational efficiency of CODE-IF compared to the other two methods. This real-data evaluation well demonstrates the real-world applicability of CODE-IF.

V. Conclusion

We have developed a state-of-the-art HSI/MSI fusion algorithm, called CODE-IF, which blends the advantages of convex analysis and deep learning. Convex analysis is known to be highly effective for the fusion mission, but related CO-based methods often rely on sophisticated regularization schemes that, however, slow down the fusion procedure (see Tables II–V). Though the deep PnP strategy has been proposed in existing literature for fast implementation of those sophisticated regularizers, the approximated proximal mapping has no convergence guarantee, in general. Thus, we are motivated to utilize the DE-based technique in a novel alternative manner, aiming at tailoring a simple DE-based convex regularizer for efficient HSI super-resolution. By combining CO-based and DE-based theories, the proposed CODE-IF algorithm does achieve some remarkable properties.

Remarkably, typical deep fusion methods rely on big data and complicated network architecture, which are often time-consuming and resource-intensive. Instead, our deep regularizer just needs a small-data-driven simple network (see Table VI) that implies better stability and tractability; this is achieved by reconsidering the role of DE-based technique as simply to guide the CO-based algorithm to search the optimal fusion solution, rather than directly using the DE-based solution to serve as the final solution. The proposed CODE-IF algorithm is theoretically rigorous and has all the closed-form expressions derived and refined to have much lower computational complexity. Even just with about 10% amount of small data, CODE-IF achieves state-of-the-art performance both quantitatively and qualitatively. This is critical because, in the upcoming space exploration era, there must be some hyperspectral scenarios (e.g., new planet exploration with unknown minerals) wherein a large amount of high-quality data is unavailable for learning the desired (but previously unknown) data statistics, implying that traditional DE theory (relying on big data) is insufficient.

Appendix

A. Proof of Property 1

What we need to show is $\text{span}(\mathbf{E}) = \text{span}(\mathbf{Y}_h)$. To this end, as (7) and (10) imply that $\mathbf{Y}_h = \mathbf{ESB}$, it suffices to verify that $\mathbf{S}$ is full row rank and that $\mathbf{B}$ is full column rank. First, from the full column rankness of $\mathbf{A}$ and the full row rankness of $\mathbf{S}'$, (9) implies that the rank of $\mathbf{Z}$ is N. Thus, (10) implies that $\mathbf{S}$ must has full row rank.

The remaining task is to show that $\mathbf{B}$ has full column rank. From [3, Eq. (34)], there exists a permutation matrix $\Pi \in \mathbb{R}^{L\times L}$ such that

$$\mathbf{B} = \Pi\left(\mathbf{I}_{L_h} \otimes \mathbf{g}\right) \in \mathbb{R}^{L\times L_h} \tag{23}$$

where $\mathbf{g}$ is a nonzero blurring column vector (i.e., vectorized convolution kernel from some Gaussian point spread function) [3]. Obviously, for any $\mathbf{g} \neq \mathbf{0}$, the matrix $\mathbf{I}_{L_h} \otimes \mathbf{g}$ has full column rank. Then, (23), together with the fact that the permutation matrix Π always has full rank, directly implies that $\mathbf{B}$ has full column rank as well. Therefore, the proof of Property 1 has been completed. ■

B. Proof of Property 2

It suffices to show that

$$\|\text{vec}(\mathbf{ET} - \mathbf{ES}_{\text{DE}})\|_{\boldsymbol{Q}}^2 = \|\mathbf{T} - \mathbf{S}_{\text{DE}}\|_F^2 \tag{24}$$

for which we define $\mathbf{t} \triangleq \text{vec}(\mathbf{T})$ and $\mathbf{s}_{\text{DE}} \triangleq \text{vec}(\mathbf{S}_{\text{DE}})$, thereby allowing us to reformulate (24) as

$$\|(\mathbf{I}_L \otimes \mathbf{E})(\mathbf{t} - \mathbf{s}_{\text{DE}})\|_{\boldsymbol{Q}}^2 = \|\mathbf{t} - \mathbf{s}_{\text{DE}}\|_2^2. \tag{25}$$

Now we just need to focus on proving (25) as follows:

$$\begin{aligned}
&\|(\mathbf{I}_L \otimes \mathbf{E})(\mathbf{t} - \mathbf{s}_{\text{DE}})\|_{\boldsymbol{Q}}^2 \\
&= [(\mathbf{I}_L \otimes \mathbf{E})(\mathbf{t} - \mathbf{s}_{\text{DE}})]^T \left[\mathbf{I}_L \otimes \left(\mathbf{EE}^T\right)\right][(\mathbf{I}_L \otimes \mathbf{E})(\mathbf{t} - \mathbf{s}_{\text{DE}})] \\
&= (\mathbf{t} - \mathbf{s}_{\text{DE}})^T \left\{\left(\mathbf{I}_L \otimes \mathbf{E}^T\right)\left[\mathbf{I}_L \otimes \left(\mathbf{EE}^T\right)\right](\mathbf{I}_L \otimes \mathbf{E})\right\}(\mathbf{t} - \mathbf{s}_{\text{DE}}) \\
&= (\mathbf{t} - \mathbf{s}_{\text{DE}})^T \left[\mathbf{I}_L \otimes \left(\mathbf{E}^T\mathbf{EE}^T\mathbf{E}\right)\right](\mathbf{t} - \mathbf{s}_{\text{DE}}) \\
&= (\mathbf{t} - \mathbf{s}_{\text{DE}})^T \ \mathbf{I}_{NL}(\mathbf{t} - \mathbf{s}_{\text{DE}}) \\
&= \|\mathbf{t} - \mathbf{s}_{\text{DE}}\|_2^2
\end{aligned} \tag{26}$$

which has completed the proof of Property 2. ■

C. Proof of Property 3

Recall the eigenspace representation of the data-fitting criterion in (12), that is,

$$\text{DF}(\mathbf{S}) = \|\mathbf{Y}_m - \mathbf{DES}\|_F^2 + \|\mathbf{Y}_h - \mathbf{ESB}\|_F^2 \tag{27}$$

which can be equivalently recast into the vectorized version $\mathbf{s} \triangleq \text{vec}(\mathbf{S})$, that is,

$$\text{DF}(\mathbf{s}) = \|\mathbf{y}_m - (\mathbf{I}_L \otimes (\mathbf{DE}))\mathbf{s}\|_2^2 + \|\mathbf{y}_h - \left(\mathbf{B}^T \otimes \mathbf{E}\right)\mathbf{s}\|_2^2 \tag{28}$$

in which $\mathbf{y}_m \triangleq \text{vec}(\mathbf{Y}_m)$ and $\mathbf{y}_h \triangleq \text{vec}(\mathbf{Y}_h)$. Define $\mathbf{y}' = [\mathbf{y}_1^T,\ \mathbf{y}_2^T]^T$ and $\mathbf{M}' = [\mathbf{M}_1^T,\ \mathbf{M}_2^T]^T$. Then, according to Pythagorean theorem, we have $\|\mathbf{y}_1 - \mathbf{M}_1\mathbf{s}\|_2^2 + \|\mathbf{y}_2 - \mathbf{M}_2\mathbf{s}\|_2^2 = \|\mathbf{y}' - \mathbf{M}'\mathbf{s}\|_2^2$, from which the proof of Property 3 directly follows from (28). ■

D. Proof of Lemma 1

We begin by observing that there exists a partition $\{\mathcal{J}_1, \ldots, \mathcal{J}_{L_h}\}$ of the index set $\mathcal{I}_L \triangleq \{1, \ldots, L\}$ such that

$$\mathcal{I}_L = \mathcal{J}_1 \cup \cdots \cup \mathcal{J}_{L_h} \tag{29}$$

$$|\mathcal{J}_n| = r^2, \ \forall n \in \mathcal{I}_{L_h} \triangleq \{1, \ldots, L_h\} \tag{30}$$

$$\mathbf{y}_h[n] = \mathbf{Z}_{\mathcal{J}_n}\mathbf{g}, \ \forall n \in \mathcal{I}_{L_h} \triangleq \{1, \ldots, L_h\} \tag{31}$$

where the hyperspectral pixel $\mathbf{y}_h[n]$ is the nth column of $\mathbf{Y}_h$; $\mathcal{J}_n$ collects the pixel indices of the target image $\mathbf{Z}$ covered by the spatial region of $\mathbf{y}_h[n]$; $\mathbf{Z}_{\mathcal{J}_n} \in \mathbb{R}^{M\times r^2}$ is the matrix sequentially formed by the r^2 columns of $\mathbf{Z}$ with column indices specified by $\mathcal{J}_n$ [3], [46]. The above observation can be physically interpreted as that, each LR hyperspectral

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pixel $\mathbf{y}_h[n]$ can be modeled by blurring the r^2 HR hyperspectral pixels in the target image $\mathbf{Z}$ covering the same spatial region of $\mathbf{y}_h[n]$. By (29), (30), and $r^2 = L/L_h$, one can see that all the $\mathcal{J}_n$'s are disjoint (i.e., $\mathcal{J}_{n_1} \cap \mathcal{J}_{n_2} = \emptyset$ for any $n_1 \neq n_2$), which, together with (31), guarantees the existence of a permutation matrix $\Pi \in \mathbb{R}^{L\times L}$ satisfying the two conditions

$$\mathbf{Z}\Pi = \left[\mathbf{Z}_{\mathcal{J}_1}, \ldots, \mathbf{Z}_{\mathcal{J}_{L_h}}\right] \tag{32}$$

$$\mathbf{Y}_h = (\mathbf{Z}\Pi)\left(\mathbf{I}_{L_h} \otimes \mathbf{g}\right). \tag{33}$$

By comparing (7) and (33), one can trickily take $\mathbf{B} = \Pi\,(\mathbf{I}_{L_h} \otimes \mathbf{g}) \in \mathbb{R}^{L\times L_h}$ and assume, without loss of generality, that pixels in $\mathbf{Z}$ have already been well arranged, that is, $\mathbf{Z} = [\mathbf{Z}_{\mathcal{J}_1}, \ldots, \mathbf{Z}_{\mathcal{J}_{L_h}}]$ and $\Pi = \mathbf{I}_L$ ($L = L_h r^2$), resulting in

$$\mathbf{B}^T \otimes \mathbf{E} = \mathbf{I}_{L_h} \otimes \left(\mathbf{g}^T \otimes \mathbf{E}\right) \tag{34}$$

$$\mathbf{I}_L \otimes (\mathbf{DE}) = \mathbf{I}_{L_h} \otimes (\mathbf{I}_{r^2} \otimes \mathbf{DE}). \tag{35}$$

By (34) and (35), one can derive

$$\begin{aligned}
\mathbf{s}^{q+1} &= \left[\mathbf{C}^T\mathbf{C} + \frac{p}{2}\mathbf{I}_{NL}\right]^{-1}\left[\mathbf{C}^T\mathbf{y} + \frac{p}{2}\boldsymbol{\psi}^q\right] \\
&= \left[\mathbf{I}_{L_h} \otimes \left(\bar{\mathbf{C}}^T\bar{\mathbf{C}}\right) + \frac{p}{2}\mathbf{I}_{NL}\right]^{-1}\left[\mathbf{C}^T\mathbf{y} + \frac{p}{2}\boldsymbol{\psi}^q\right] \\
&= \left[\mathbf{I}_{L_h} \otimes \left(\bar{\mathbf{C}}^T\bar{\mathbf{C}} + \frac{p}{2}\mathbf{I}_{Nr^2}\right)\right]^{-1}\left[\mathbf{C}^T\mathbf{y} + \frac{p}{2}\boldsymbol{\psi}^q\right] \\
&= \left[\mathbf{I}_{L_h} \otimes \left(\bar{\mathbf{C}}^T\bar{\mathbf{C}} + \frac{p}{2}\mathbf{I}_{Nr^2}\right)^{-1}\right]\left[\mathbf{C}^T\mathbf{y} + \frac{p}{2}\boldsymbol{\psi}^q\right].
\end{aligned} \tag{36}$$

Note that in (36), we just need to compute the inverse of an $(Nr^2) \times (Nr^2)$ matrix, much faster than the $(NL) \times (NL)$ matrix inversion in (20). Nevertheless, computing $\mathbf{C}^T\mathbf{y}$ is still time-consuming. Although separately computing the term $\mathbf{C}^T\mathbf{y}$ induces a high complexity, a tricky approach is to directly compute the first part of (20), that is, $(\mathbf{C}^T\mathbf{C} + (p/2)\,\mathbf{I}_{NL})^{-1}\mathbf{C}^T\mathbf{y}$; to be precise, one can apply (34) and (35) to decouple the term as follows:

$$\begin{aligned}
&\left(\mathbf{C}^T\mathbf{C} + (p/2)\,\mathbf{I}_{NL}\right)^{-1}\mathbf{C}^T\mathbf{y} \\
&= \left(\mathbf{C}^T\mathbf{C} + (p/2)\,\mathbf{I}_{NL}\right)^{-1}\left[\mathbf{I}_{L_h} \otimes \left(\mathbf{g}^T \otimes \mathbf{E}\right)^T\right]\mathbf{y}_h \\
&\quad + \left(\mathbf{C}^T\mathbf{C} + (p/2)\,\mathbf{I}_{NL}\right)^{-1}\left[\mathbf{I}_{L_h} \otimes (\mathbf{I}_{r^2} \otimes \mathbf{DE})^T\right]\mathbf{y}_m \\
&= \left\{\mathbf{I}_{L_h} \otimes \left[\left(\bar{\mathbf{C}}^T\bar{\mathbf{C}} + (p/2)\,\mathbf{I}_{Nr^2}\right)^{-1}\left(\mathbf{g}^T \otimes \mathbf{E}\right)^T\right]\right\}\mathbf{y}_h \\
&\quad + \left\{\mathbf{I}_{L_h} \otimes \left[\left(\bar{\mathbf{C}}^T\bar{\mathbf{C}} + (p/2)\,\mathbf{I}_{Nr^2}\right)^{-1}(\mathbf{I}_{r^2} \otimes \mathbf{DE})^T\right]\right\}\mathbf{y}_m
\end{aligned} \tag{37}$$

which, together with (36), directly yields (21).

The remaining task is to derive the complexity in Lemma 1. Remarkably, as can be seen from the above derivation, the first part of (20) is a constant term that just needs to be computed once. For the first term in (21), the complexity is dominated by the three parts.

1) The computation of $\bar{\mathbf{C}}^T\bar{\mathbf{C}}$ requires a complexity of $\mathcal{O}((Nr^2)^2(M + r^2M_m))$.
2) The computation of the (Nr^2)-dimensional matrix inversion takes $\mathcal{O}((Nr^2)^3)$.
3) The computation of the L_h matrix–vector multiplications, involving an $(Nr^2) \times (Nr^2)$ matrix and an (Nr^2)-dimensional vector, takes $\mathcal{O}(L_h(Nr^2)^2)$.

For the second term in (21), the complexity is dominated by the four parts.

1) The computation of $\bar{\mathbf{C}}^T\bar{\mathbf{C}}$ takes a complexity of $\mathcal{O}((Nr^2)^2(M + r^2M_m))$.
2) The computation of (Nr^2)-dimensional matrix inversion takes $\mathcal{O}((Nr^2)^3)$.
3) The computation of multiplying an $(Nr^2)\times(Nr^2)$ matrix by an $(Nr^2) \times M$ matrix takes $\mathcal{O}((Nr^2)^2M)$.
4) The computation of the L_h matrix–vector multiplications, involving an $(Nr^2) \times M$ matrix and an M-dimensional vector, takes $\mathcal{O}(L_hNr^2M)$.

For the third term in (21), the complexity is dominated by the four parts.

1) The computation of the core term $\bar{\mathbf{C}}^T\bar{\mathbf{C}}$ takes $\mathcal{O}((Nr^2)^2(M + r^2M_m))$.
2) The computation of the (Nr^2)-dimensional matrix inversion takes $\mathcal{O}((Nr^2)^3)$.
3) The computation of multiplying an $(Nr^2)\times(Nr^2)$ matrix by an $(Nr^2) \times (M_mr^2)$ matrix takes $\mathcal{O}((Nr^2)^2M_mr^2)$.
4) The computation of the L_h matrix–vector multiplications, involving an $(Nr^2) \times (M_mr^2)$ matrix and an (M_mr^2)-dimensional vector, takes $\mathcal{O}(L_hNr^2M_mr^2)$.

Therefore, by defining $\xi \triangleq \max\{M, M_mr^2, Nr^2\}$, the overall computational complexity can be simplified as $\mathcal{O}(Nr^2(Nr^2 + L_h)\xi) = \mathcal{O}((NL+(Nr^2)^2)\xi)$, and hence the proof of Lemma 1 has been done. ■

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